

1)Which is the current version you are working on in ServiceNow?

Answer)Washington DC-latest one

2)From which version you started working

Answer)Rome, san diego,tokyo, utha,vancouver,Washington Dc

3)Can we add permissions to the users and groups? which is the best practice?

Answer)Yes , we can add roles to the users and groups manually and as well as with the script by using GlideRecord. the best practice is to add the permissions from group...so that whenever an employee leaves an organisation and if we remove that person from the group the roles will be removed automatically.

4)What is the user table name?

Answer)sys_user

5)What is the group member table name?

Answer)sys_user_group

6)How to create user account using script?

Answer)By using glide record as follows

```
var userGr = new GlideRecord('sys_user');
userGr.initialize();
userGr.username='jdoe';
userGr.firstname='John';
userGr.lastName = 'Doe';
userGr.email = 'jdoe@example.com';
userGr.insert();
```

7)How to create group using script?

Answer)By using GlideRecord as follows

```
var newGr = new GlideRecord('sys_user_group');
newGr.initialize();
newGr.name='testing';
newGr.manager='62826bf03710200044e0bfc8bcbe5df1';
newGr.email='testing@tcs.com';
newGr.description='test';
newGr.insert();
```

8)How to add permissions to user/group account using script?

Answer)When ever a permission is added to a user account a record in sys_user_has_role table will get created, we can add a role to a user by using script as follows

```
var userRole=new GlideRecord('sys_user_has_role');
userRole.setValue('user','62826bf03710200044e0bfc8bcbe5df1');
userRole.setValue('role','2831a114c611228501d4ea6c309d626d');
userRole.insert();
```

--->Now for group:When we add a permission a group ,a record will be created inside sys_group_has_role table, we can add a role to a group by using script as follows

```
var grpRole=new GlideRecord('sys_group_has_role');
grpRole.setValue('group','477a05d153013010b846ddeeff7b1225');
grpRole.setValue('role','2831a114c611228501d4ea6c309d626d');
grpRole.insert();
```

9)What exactly user delegation means in ServiceNow?

Answer)User delegation means allowing a user to work on behalf of another user within an organization. This is generally used when the original user/employee is unavailable, such as when they are on vacation or leave. The delegated user gets the permissions to perform tasks and he can also get access to the resources which are normally available to the original user.

For example, in an office, if an employee is going on vacation, they can delegate their approval tasks to a colleague. This way, important tasks and workflows continue smoothly without interruption.

for this you need to go to ->original user account->scroll down->click delegates-> here you can provide the details such as delegate(name of the person ou want to delegate your work),start date, end date and you can also give the permissions for assignments, notifications , approvals.

10)How to add and remove group member from group using script?

Adding Group Member:

```
var grMem=new GlideRecord(sys_user_grmember);
grMem.user='62826bf03710200044e0bfc8bcbe5df1';
grMem.group='477a05d153013010b846ddeeff7b1225';
grMem.insert();
```

Removing Group Member:

```
var grMem=new GlideRecord(sys_user_grmember);
grMem.addQuery('user','62826bf03710200044e0bfc8bcbe5df1');
grMem.addQuery('group','477a05d153013010b846ddeeff7b1225');
grMem.query();
```

```
if(grMem.next()){  
grMem.deleteRecord();  
}
```

11)How many user interfaces are there in snow?

Answer)UI15,16 next experience

12)What is mean by web services user in the User account?

Ans)when a web services access is grant only to a 3rd party application to get connected to the ServiceNow then it is called as web services user and moreover this user can not login as a snow user into servicenow .

13)How to get the current logged in user system id in the client side?

Answer)g_user.UserID;
//6816f79cc0a8016401c5a33be04be441

14)How to get the current logged in user system id in the server side?

Answer)gs.getUserID();

15)How to check if the current logged user is the member of particular group or not?

Answer)gs.getUser.isMemberOf('group name'); //true if part of the specified group
//false if not part of specified group

16)Which role is required to work on access control?

Answer)Security_admin

17)What is impersonation?

Answer) logging in as another user for testing.

18)What is user preference?

users can change their preferences based on their wish, so when user change their preferences it will only impact to the same user it will not impact globally

19)What is incident?

Sudden interruption in the service when the employee works in the organization if something suddenly stopped working he will get the support from support engineer by creating a incident.

20)What is problem?

Answer)if the same issue is repeatedly happening to the same employee then it is called problem. if the same problem is happening to the multiple people at the same time then its an incident, where will create a parent incident and rest of all will be child incidents, whenever you close the parent incident the child incidents will be also get closed

21)Can we create problem record from incident?

Answer)yes, if the issue is repeatedly occurring then we will create a problem from incident.

22)Can We create a change request from incident?

Yes, when ever you create an incident if the support engineer feels that their should be some change in the software then he will arise a change request from that incident.

23)How to create incident record using script?

Answer)Using GlideRecord as follows

```
var gr = new GlideRecord('incident');
gr.initialize();
gr.caller_id = '86826bf03710200044e0bfc8bcbe5d94';
gr.category = 'inquiry';
gr.subcategory = 'antivirus';
gr.cmdb_ci = 'affd3c8437201000deeabfc8bcbe5dc3';
gr.short_description = 'test record using script';
gr.description = 'test record using script';
gr.assignment_group = 'a715cd759f2002002920bde8132e7018';
gr.insert();
```

24)How to create problem record using script?

Answer)Using GlideRecord as follows

```
var gr = new GlideRecord('problem');
gr.initialize();
gr.caller_id = '86826bf03710200044e0bfc8bcbe5d94';
gr.category = 'inquiry';
gr.subcategory = 'antivirus';
gr.cmdb_ci = 'affd3c8437201000deeabfc8bcbe5dc3';
gr.short_description = 'test record using script';
gr.description = 'test record using script';
gr.assignment_group = 'a715cd759f2002002920bde8132e7018';
gr.insert();
```

-----.
25)How to create change request using script?

Answer)Using GlideRecord as follows

```
var gr = new GlideRecord('change_request');
gr.initialize();
gr.category = 'inquiry';
gr.subcategory = 'antivirus';
gr.cmdb_ci = 'affd3c8437201000deeabfc8bcbe5dc3';
gr.short_description = 'test record using script';
gr.description = 'test record using script';
gr.assignment_group = 'a715cd759f2002002920bde8132e7018';
gr.insert();
```

26)Write a logic for whenever a parent incident is get closed all the child incidents should also get closed.?

Answer: Write a after business Rule

When -- After

Update - true

Condition: current.state.changesTo(7);

```
if (current.state == 7 && current.parent == "") {

    // GlideRecord to find child incidents
    var grChild = new GlideRecord('incident');
    grChild.addQuery('parent', current.sys_id);
    grChild.query();

    while (grChild.next()) {
        grChild.state = 7; // Set the state to Closed
        grChild.update(); // Update the child incident
    }
}
```

27)There is an incident and that incident has associated 2 tasks ,when u try to close that incident if any incident task is open so employee should not close the incident. Similarly for problem, and change request.?

Answer:

```
var grTask = new GlideRecord('incident_task');
grTask.addQuery('incident', current.sys_id);
grTask.addQuery('state', '!=', 3); // Assuming 3 is the state value for 'Closed'
grTask.query();

if (grTask.hasNext()) {
    gs.addErrorMessage('Cannot close the incident because there are open tasks.');
```

28)Whenever problem is closed the associated incident will also get closed.?

Answer:

```
if (current.state == 7) {

    // GlideRecord to find incidents associated with the problem
    var grIncident = new GlideRecord('incident');
    grIncident.addQuery('problem_id', current.sys_id);
    grIncident.addQuery('state', '!=', 7); // Assuming 7 is the state value for 'Closed'
    grIncident.query();

    while (grIncident.next()) {
        grIncident.state = 7; // Set the state to Closed
        grIncident.update(); // Update the incident
    }
}
```

29)What is the relationship between incident , problem and change management?

Answer:

if a person face some issue he will create an incident and if the same issue is happening again and again then he will create a problem , and if the support team feels like some changes are required in their software then they will create a change request.

30)Give me some names of out of the box tables?

Answer)the tables which doesn't starts with x_ or u_ those tables we called as out of the box, incident, x_ means id the table is create in scoped application.

31)What are some of the base tables?

Answer)task ,cmdb_ci table the tables which don't extend any table but extended by many tables.

32) give me some examples of tasks tables?

Answer:
incident, problem change request which are extending task table.

33)whenever we create a table how many access controls will get created?

Answer: 4 ACL when creating a table by default 4 AC will be created because of the check box is checked if you uncheck it will not create ACL.

34)How to create number field, like incident number?

Answer)You have to go to the control tab and u have to provide the prefix and no of digits we have to check the auto number field when creating incident table.

35) what happens when you extend a table?

Answer) when u extend a table the sys fields will not get created in child table bcz those are coming from parent table and a field called class will be created in parent table and if a table is extending many tables it will have only one class field it will not create class field for each new child table.

36)Can you give me 10 field types?

Answer)Reference,String,List,Choice,email,date/time,date,boolean,integer,journal,attachment

37)What is the difference between temporary and normal table ?

Answer) temp tables will store data temp for 7 days and they will extend import set row table. normal tables will have the data permanently

38)Can we increase the retention period in the temp table?

Answer) yes by using archive rules

39)what is the difference b/w remote table and normal tables?

Answer)remote table means you will get the live data but in the normal table u will have the data what u have stored.

40)what is one to one and one to many table inServiceNow?

Answer:

One-to-Many Relationship: Users and Incidents, where one user can have multiple incidents, but each incident is assigned to one user.

Many-to-Many Relationship: Incidents and Groups, where an incident can be associated with multiple groups, and a group can be associated with multiple incidents.

41) In how many ways we can create a record in snow table?

Answer) record producer, email, GlideRecord, form and excel sheet, from external system as well

42) In how many ways we can make a field mandatory, read only?

Answer) from dictionary properties, dictionary overrides, UI policies, data policies, and `g_form.setMandatory(Script)`.

43) Can we make 2 fields as display in one table?

Answer) No we should not make 2 fields display because it will lead to the confusion.

44) All tables will be stored where?

Answer) `sys_db_object` table.

45) How to set a default value on a field?

Answer) whenever the form is open if you want to populate some default values on the form then we can set default values for that particular field.

46) What is current object?

Answer) it is used to set and get the values on the form at the server side.

47) How to set the field values on the current form?

Answer) `current.setValue('field_name', value)`; if we are writing this for reference type of field then we should give `sys_id` in place of value, `current.setDisplayValue('field_name', value)`; here if reference field give value as value only not `sys_id`.

48) What are reference qualifiers, and their types? And explain each one in detail and what is the difference in simple and dynamic, dynamic and advanced, simple and advanced.

Answer) Reference Qualifiers are used to restrict the data in reference and List type of fields

3 Types of Reference Qualifiers:

Simple
Dynamic
Advanced

Simple Reference Qualifier:

Description: This is the most basic form of a reference qualifier where you specify a fixed query to filter the referenced records.

Example: If you want to display only active users in a reference field for the User table, you can set a simple reference qualifier.

How to Use:

In the reference field dictionary entry, set the reference qualifier condition.

Example: active=true

Dynamic Reference Qualifier:

Description: This type uses a dynamically generated query that can adapt based on the context, such as the values of other fields on the form.

Example: Displaying only incidents assigned to the same assignment group as the current user.

How to Use:

Create a dynamic filter option via System Definition > Dynamic Filter Options.

Define the conditions and make it available for use in reference qualifiers.

Use it in the reference field dictionary entry by selecting the dynamic filter option.

Advanced Reference Qualifier (JavaScript Reference Qualifier):

Description: This type uses a custom JavaScript code to define complex queries for the reference field. It allows for dynamic filtering based on multiple conditions and the current context.

Example: Filtering the Incident table to show only incidents assigned to the current user's assignment group and with a priority less than 3.

How to Use:

In the reference field dictionary entry, select Advanced for the reference qualifier and enter JavaScript code.

```
javascript: 'assignment_group=uoewuroewutoewutoewutowuot^priority<3';
```

49)What is depended value?

Answer:

Dependent Values in Dictionary

****Dependent values**** in a dictionary entry in ServiceNow are used to filter the available choices in one field based on the selection made in another field. This is commonly used for creating cascaded dropdown menus.

Steps to Configure Dependent Values

1. ****Identify Parent and Dependent Fields****:
 - Determine which field will be the parent (e.g., Category) and which will be the dependent (e.g., Subcategory).
2. ****Configure the Parent Field****:
 - Ensure the parent field has a list of possible values.
3. ****Configure the Dependent Field****:
 - Go to the dictionary entry of the dependent field.
 - Set the ****Dependent Field**** attribute to the parent field.
 - Define choices for the dependent field linked to specific values of the parent field.

Example

1. ****Parent Field****: Category (values: Hardware, Software, Network)
2. ****Dependent Field****: Subcategory (dependent on Category)

Define choices for Subcategory:

- If Category = Hardware: Laptop, Desktop, Printer
- If Category = Software: Operating System, Application, Database
- If Category = Network: Router, Switch, Firewall

By configuring this, the Subcategory dropdown dynamically updates based on the Category selection, showing only relevant options.

50)What is calculated value?

Answer: If we want to calculate a field value based on other field value using current object, we use calculated Dictionary Property.

51)What are attributes, name some of the attributes that you used?

Answer: Attributes are used to change the field behavior on dictionarys(Fields) Ex. no_email, no_attachment, no_add_me, tree_picker..Etc

52)What is collection field on table?

Answer:Dictionary entries with the Collection type represent the table rather than a field on the table. Changes such as attributes or the Read only check box applied to this entry are applied to the table rather than a specific field. This entry is automatically created when a table is created.

53)How to enable/disable attachment on the form using attributes?

Answer) In the collection field u will write no_attachment attribute

54)What are dictionary overrides? what are all the properties we can override in dictionary overrides?

Answer:

Use a dictionary override to allow a field in a child table to have a different value or behavior than the same field in a parent table. For example, a dictionary override changes the default value of the priority field from 4 in the parent table to 5 in the Incident table.

55)What are application menus?

Answer) To make the forms and list available to end user we can create a module in the snow using application menus.

56)What is process flow?

Answer) Process flow is an indication to show the state of the form on which we are.

57)What are data lookup roles?

Answer)To populate the field values based on the field values on the form its a separate table

58)What are ui policies?

Answer)UI policies are used to make a field mandatory, read only ,display and to show the related lists at certain condition in the client side.

59)what is global check box in ui policies?

Answer)When this box is checked then ui policy will work on all the views, when we uncheck it will ask us to give the view name then in this case the ui policy will work only on that specific view.

60)what is reverse if false in ui policies?

Answer)If the "Reverse if false" checkbox is selected, the actions applied by the UI Policy will be reversed when the condition evaluates to false. For example, if a field was made mandatory when the condition was true, it will become optional again when the condition is false.

61)What is onload check box in ui policy?

Answer)When the "On Load" checkbox is selected, the conditions and actions specified in the UI Policy are evaluated and applied immediately when the form is loaded.If the "On Load" checkbox is not selected, the UI Policy actions will not be applied when the form is initially loaded. Instead, the actions will only be triggered when a specific field or condition is meet on the form.

62)what is inherit checkbox ?

Answer)When this check box is checked the same ui policy will be applied to child tables as well, means the table which are extending the table on which we applied ui policy.

63)can you write script in ui policy?

Answer)Yes, you can write scripts in a UI Policy in ServiceNow, For adding a script, you will need to enable Run scripts checkbox.

Once you have created a UI Policy Action, enable the Run scripts checkbox. This will allow you to enter a script that will execute when the UI Policy condition is met.

64)can we convert ui policy as data policy?

Answer)yes we can convert ui policy into data policy for that you need to open the data policy and click on "convert this as data policy".

65)Which are all the cases where ui policy cannot be converted as data policy?

Answer)When u r controlling the data visibility, When you are controlling the views, When you are controlling the related lists, When you are controlling the script.

66)what are data policies?

Answer)Data policies are used to make field mandatory, read only, display at certain condition from all the data sources ,and this works at both client and server side.

67)what is the diff b/w ui and data policies?

Answer)

- 1.UI policy works at client side and data policy works at both client and server side.
 - 2.You can't hide a field on the form using data policy
 - 3.In data policy we can't write the script
 - 4.You can not make data policy to a specific view and UI policy can be applied for a particular view
 - 5.Data policy works on all the sources, where as ui policy is only works on the form.
-

68)What is Glid user api?

Answer)It's used to get the current logged in user information and their roles at client side. ands it can be accessed using g_user.

69)name some of the methods in the glide user api?

Answer)

```
g_user.firstName;  
g_user.lastName;  
g_user.getFullName();  
g_user.userName;  
g_user.userID;
```

70)what is glide form api?

Answer)It is used to change(or) make the modification on the form at client side and it can be accessed using g_form.

71)name some of the methods in the glide form api?

Answer)g_form.addInfoMessage("");
g_form.setMandatory("field name","true/false");
g_form.setReadOnly("field name","true/false");
g_form.enableAttachments("");
g_form.addErrorMessage("");
g_form.addOption("field_name","field_value");

72)What are client scripts and how many types of client scripts explain about each?

Answer)Client scripts are a piece of java scripts which get executed at client side, there are 4 types of client scripts

1)OnLoad: It will run every time when the form gets loaded.

2)OnChange:It will get executed when you change a value on a field , And it will also get executed when form gets loaded.

3)OnSubmit:This will be executed before form submission.

4)OnCellEdit:it will not work on the form and will be run when you change a value on a particular cell on the list

73)Which one runs first client script or ui policy?

Answer)client script will run first and UI policy will run at last.

74)What are the onChange client script parameters?

Answer)They are as follows:

1.control: This is used to get the HTML element of the field on which script is applied.

2.oldValue:used to get the old value of the field

3.newValue:used to get the new value of the field

4.isLoading:When the form is loading this will give Boolean value (true/false)

5.isTemplate:We use this parameter when we are not filling the form manually but using template

75)can we make on Change client script to work on load client script?

Answer)yes, we can make the onChange client script to work on the form when it gets loaded by removing the following condition from the script

```
if (isLoading || newValue === '') {  
    return;  
}
```

76)how to stop form submission using on submit?

Answer: return false;

78)What are the oncellEdit script parameters?

Answer)They are 5 as follows:

1.sysIDs:
2)Table
3)oldValues
4)NewValue
5)callback

79)ui actions client side or server side?

Answer)By default on server side but can be work at client side by checking the cline check box in UI action

80)How to move from client ui actions to server side ui actions?

Answer: using a function, gstSubmit(null, g_form.getFormElement(), 'action_name');

81)What is glideAjax?

Answer)used to get the data from server database to client side , its a client side API

82)What is glideAjax syntax?

Answer)with asynchronous

Syntax:

```
var ga = new GlideAjax('YourScriptInclude');  
ga.addParam('sysparm_name', 'yourFunction');  
ga.addParam('sysparm_caller', g_form.getValue('u_caller'))  
ga.getXMLAnswer(callbackFunction);
```

```
function callbackFunction(response) {  
    var answer = response;  
    // Process the response  
}
```

83)Example of glide ajax?

Answer: GlideAjax is used to execute server script from client script

You can pass values as inputs in client script and pass it to server script, from server script, you can pass output as response to client scripts in servicenow.

84)How many types of glideajax?

Answer)Asyn GlideAjax and Sync GlideAjax

85)what is synchronous and asynchronous?

Answer: Synchronous getReference() method calls lock the form until the requested record is returned from the database. Asynchronous execution allows users to continue using the form while the method call executes. The first example executes synchronously resulting in a poor user experience.

86)example for synchronous glidajax?

Answer: When caller selected is Contractor, The form submission should be stopped. Then we will use synchronous GlideAjax.

```
var ga = new GlideAjax('HelloWorld') ;
ga.addParam('sysparm_name','helloWorld');
ga.addParam('sysparm_user_name','Bob");
ga.getXMLWait();
alert(ga.getAnswer());
```

Synchronous calls can be useful in situations where you need to ensure the server response is processed before allowing the user to continue interacting with the page. However, they should be used sparingly due to the potential negative impact on performance and user experience. In most cases, asynchronous calls are preferred.

87)What is glide navigation api?

navigation api can be accessed using g_navigation its used to navigate the urls in diff tab this can be used only at the client side.

88)in how many ways we can get the values from server to client side?

Answer)4 ways....gliderecord,glideajax,g_form.getReference and scratch pad.

89)What are ui pages?

Answer)ui pages are client side script,they are like web pages which are build with html css and jelly script.

90)can we call ui pages in the client script,ui policies and client side ui actions?

Answer: Yes we can call.

91)how to call ui pages in the client side script?

Answer:
using GlideDialogWindow or GlideModal API

```
var gdw = new GlideModal('ui_page_name');
gdw.setTitle('Resolution Information');
gdw.setSize(300,400);
gdw.setPreference('keyid', g_form.getUniqueValue())
gdw.render();
```

92)How to pass values from client side script to ui pages?

Answer: Using a method in GlideModal, That is setPreference method.

```
var gdw = new GlideModal('ui_page_name');
gdw.setTitle('Resolution Information');
gdw.setSize(300,400);
gdw.setPreference('keyid', g_form.getUniqueValue())
gdw.render();
```

93)How do you access values in the ui page from the client script?

Answer:<g:evaluate var="jvar_text" expression ="RP.getWindowProperties().get('text')">

94)what is processing script in ui page?

Answer: Processing script is server side script. the processing script will be executed when we write the script in <g:ui_form></g:ui_form>

95)How to close ui page automatically using script?

Answer: GlideDialogWindow.get().destroy();

96)what are ui macros?

Answer)UI macros are basically used to create one's own custom controls and interfaces. Administrator can create UI Macros. For creating UI Macros, knowledge of jelly scripting is required. By default, ServiceNow provides UI Macros for various user interfaces like: All formatters.To create custom field on the form like radio buttons, to make advance customizations on the form we use ui macros.

97)How to call ui macro on the form?

Answer: using UI formatters, in the UI formatter, we have to call UI Macro first, using macro_name.xml then we have to add formatter on the form.

98)how to call ui macro on the dictionary?

Answer: In the Dictionary, In the attribute column, We have to add a attribute called ref_contributions=macro_name

99)How to hide ui macro using script?

Answer: We have to use JavaScript DOM

Hiding UI Macro: document.getElementById('uimacro_id').style.display = 'none'

Showing UI Macro: document.getElementById('uimacro_id').style.display = 'block'

We need to make sure the Isolate script check box is unchecked.

100)can we call ui page from uimacro?

Answer: Create a button tag in UI Macro. In the button tag, add a onclick attribute, when we click on it, it will call the function, we have to keep the function inside script tag.

we have to write below script to call UI pages in UI macro.

```
var gdw = new GlideModal('ui_page_name');  
gdw.setTitle('Resolution Information');  
gdw.setSize(300,400);  
gdw.setPreference('keyid', g_form.getUniqueValue())  
gdw.render();
```

101)what is isolate checkbox in client script?

Answer)it is a setting that controls whether the script runs in its own isolated environment or not. When this box is checked, it means that the script executes independently without being influenced by the changes which are made by other scripts, and it will also not affect other scripts. It's useful for ensuring that the script's behavior is predictable and not influenced by external factors and to maintain script reliability and predictability.

102)what are the best practices in the client side?

Answer)

Few of the best practices to use client Scripts :

Enclose Code in Functions.
Avoid DOM manipulation, use g_form object.
Avoid global client scripting, etc.
Not using g_form.getReference.
Not using too many alerts.
Using asynchronous calls in client side

103)What are views in ServiceNow?

Answer)A view defines the elements that appear when a user opens a form or a list, and you can switch the view from the default for lists and forms.one form can be shown in multiple views.

104)How do you control the views?

Answer)By using view rules.

105)Syntax for view rule?(confusion)

Answer)

```
var dept = gs.getUser().getRecord().getDisplayValue('department');
```

```
if (dept == 'Development') {  
    answer = 'Development';  
} else if (dept == 'Sales') {  
    answer = 'Sales';  
} else if (dept == 'HR') {  
    answer = 'Human_Resource';  
} else if (dept == 'Finance') {  
    answer = 'Finance';  
} else {  
    answer = 'default_view';  
}
```

106)What is the diff b/w setVisible and setDisplay?

Answer)The difference between setVisible and setDisplay is that, if we are trying to hide a field using setVisible as setVisible('u_assignment_group',false);, then it will hide that field from the form but the space of that field will not removed it will remain same , so to hide the field along with the space on the form we can use setDisplay, as setDisplay('u_assignment_group',false).

107)How to hide all sections using script?

```
Answer)function onLoad() {  
    // Get all section names  
    var sections = g_form.getSectionNames();  
  
    // Loop through each section and hide it  
    for (var i = 0; i < sections.length; i++) {  
        g_form.setSectionDisplay(sections[i], false);  
    }  
}
```

108)How to make all the field read only which are editable?

```
var field = g_form.getEditableFields();  
  
for (var i = 0; i < field.length; i++) {  
    g_form.setReadOnly(field[i], true);  
}
```

109)How to hide all the related lists at the same time?

```
Answer)  
g_form.hideRelatedLists();
```

110)How to show all the sections?

```
function onLoad() {  
    var sections = g_form.getSectionNames();  
  
    for (var i = 0; i < sections.length; i++) {  
        g_form.setSectionDisplay(sections[i], true);  
    }  
}
```

111)How to hide new and edit button in the related list to certain people ?

```
Answer)in the related list click on those 3 dots(Column Options) and click on that omit(need to check practically)
```

112)How to make the attachment mandatory on the form before the form submission?

Answer):

Simply create a BEFORE UPDATE/INSERT Business rule and add following code:

```
(function executeRule(current, previous /*null when async*/) {  
  
gs.addInfoMessage("Result:"+current.hasAttachments());  
  
if(current.hasAttachments() != true){  
  
gs.addInfoMessage("PLease add an attachment");  
  
current.setAbortAction(true);  
  
}
```

113)How do we restrict assigned to based on the assignment group?

Answer)using advanced ref qualifies

114)What are update sets?

Answer)Update sets are used to move your work from one environment(DEV) to another environment(TEST).

115)how many ways we can move update set to the target instance?

Answer)

Export XML and Import XML

Update source.

Transferring with IP access control.

116)How do u migrate the update set from one instance to another instance?

Firstly we have to update set in source instance, then export the xml or import the xml in target instance second way is retrieving the update set using retrieved update set from update set source .

Preview the update set and commit the update set

117)How do u handle the preview errors?

Answer)By using "skip remote update" and "accept remote update". means when we want to keep the changes made in the other environment then we have to do accept remote update and if we don't want to keep the changes made in the changes then do the "skip remote update". so this is how we handle preview errors in the update set.

118) Which configurations get captured in the update sets and which are not get captured?

Answer)

the update set captures the system configuration changes you make to the system. This includes, but is not limited to:

Tables
fields
form layout
list layout
business rules, client scripts, script includes, and other scripts
workflows
system properties

What it does NOT capture is your transactional data like

Incidents
Users
Groups

In ServiceNow everything will be stored in tables, so the tables which are having attribute called 'update_synch=true' then those tables or modules will get captured in the update set, and the tables which are not having this attributes will not get captured.

119) Can we move an update set to in progress which is already migrated to another instance?

Answer:

We should not move the update set which is already migrated to another instance, instead, we have to create a new update set with different version.

120) Can we capture data in an update set using script ?

Answer:

```
var um = new GlideUpdateManager2();  
um.saveRecord(object);
```

121) What happens when u close the default update set?

Answer) a new default update set will be created as 'default1'.

122) What is merge update set?

Answer) merge update set in ServiceNow is a process that combines multiple update sets into a single update set u close that update set and moved it to the target instance as one update set, This is useful when you have several update sets that contain related changes, and you want to merge them into one update set for easier management and deployment

123)What is batch update set?

Answer)first u create update set make one update set as parent. once you close the parent update set child update set will be closed. once u move the batch update set to target instance all the child update sets will also be moved automatically, once you preview the update set the child update sets will be previewed auto. and once you commit the batch update set child update sets will be committed automatically

124)What is GlideModal in the client side?

Answer: GlideModal is used to call UI pages in the client Side.

125)what is GlideModalForm in client side?

Answer: GlideModalForm is is used to call Table forms from client side such as Client side UI actions, Client Scripts, UI Policys, UI Macros.

126)What is the difference between getXML and getXMLwait ?

Answer)getXML is asynchronous meaning it sends a request to the server and continues executing the rest of the script without waiting for the response. When the response is received, a specified callback function is called to handle the response.It is used when you do not want to block the user interface while waiting for a response from the server.

Syntax:

```
var ga = new GlideAjax('YourScriptInclude');
ga.addParam('sysparm_name', 'yourFunction');
ga.addParam('sysparm_caller', g_form.getValue('u_caller'))
ga.getXMLAnswer(callbackFunction);
```

```
function callbackFunction(response) {
    var answer = response;
    // Process the response
}
```

And getXMLWait is a synchronous method, meaning it sends a request to the server and waits for the response before continuing to execute the rest of the script. This method blocks the execution of the script until the server responds.

It is used when the subsequent code execution depends on the server response, and you need to wait for the response before proceeding. However, it can make the user interface unresponsive if the server takes time to respond.

Syntax:

```
var ga = new GlideAjax('YourScriptInclude');
ga.addParam('sysparm_name', 'yourFunction');
ga.getXMLWait();
var answer = ga.getAnswer();
// Process the response
```

127)What is backout update update set?

Answer)Once you capture the development in update set and then u realised that that development is not required in the end u can click on backout in the update set and then all you captured data will get erased

128)What update_synch attribute in servicenow?

Answer: In ServiceNow everything will be stored in tables, so the tables which are having attribute called 'update_synch=true' then those tables or modules will get captured in the update set, and the tables which are not having this attributes will not get captured.

129)List of Client Side APIs:

GlideUser
GlideForm
GlideAjax
GlideModal
GlideModalForm
GlideDialogWindow
GlideList
GlideNavigation
GlideScratchpad

130)List of Server Side APIs:

GlideRecord
GlideRecordSecure
GlideAggregate
GlideDateTime
GlideDate
GlideSchedule
GlideDuration
GlideUpdateManager2
GlideImportTransformer
GlideExcelParser

131) Can we use GlideRecord in client scripts?

Answer)We can use GlideRecord in client Script, But using GlideRecord in Client Script cause performance issue. It is not good practice to use GlideRecord in client script.

132)What is the purpose of GlideRecord?

Answer) GlideRecord is used for CRUD operations on tables in ServiceNow database.

134) What is the difference between GlideRecord and GlideRecordSecure?

Answer) GlideRecord and GlideRecordSecure both are used for CRUD operations, GlideRecordSecure by default checks the permissions, Whereas Using GlideRecord, we have to specifically check permissions using canCreate(), canWrite(), canDelete(), canRead().

135) How to set a field unique on the table?

Ans: Go to that respective field dictionary and set the unique check box to true.

136) What is the setWorkflow(false) function does?

Ans: setWorkflow(e) enables or disables the running of business rules that might normally be triggered by subsequent actions. If the e parameter is set to false, an insert/update will not be audited. Auditing only happens when the parameter is set to true for a GlideRecord operation.

Parameters:

e – A boolean variable that if true (default) enables business rules, and if false to disables them.

137) What the setForceUpdate() function does?

Ans: setForceUpdate() updates the record even if there are no changes on the record.

```
//Force an update to all User records without changing field values
var gr = new GlideRecord('sys_user');
gr.query();
while (gr.next()) {
gr.setWorkflow(false); //Do not run business rules
gr.autoSysFields(false); //Do not update system fields
gr.setForceUpdate(true); //Force the update
gr.update();
}
```

138) What is the significance of the set limit(n) function?

Ans: set limit(n) functions to limit the number of records to query by Gliderecord().

139) How to check database operations using script in Business Rules?

Answer: current.operation();

140) How to stop data processing using Before Business Rules in ServiceNow?

Answer: current.setAbortAction(true);

141) Can we use current.update() in Before Business Rules?

Answer: we should not use `current.update()` in Before Business Rules.

142) Can we use `current.update()` in after business rules?

Answer: using `current.update()` in after business rule is not good practice, if you want to use, we have to use `current.setWorkflow(false)` before using `current.update()`;

```
current.setWorkflow(false);  
current.update();
```

143) Can you update a record without updating its system fields(like `sys_updated_by`, `sys_updated_on`)?

Ans: Yes, you can do it by using a function `autoSysFields()` in your server-side scripting. Whenever you are updating a record set the `autoSysFields()` to false.
example:

```
var gr = new GlideRecord('incident');  
gr.query();  
if(gr.next()){  
gr.autoSysFields(false);  
short_description = "Test from Examsmynta" ;  
gr.update();  
}
```

144) What is the difference between after business rule and Async Business rule?

Answer: After business Rules run synchronously, Async business Rules run asynchronously. After BRs will wait for the complete script execution and it will cause performance issue. Async BRs will run asynchronously and it will run in the background using a system scheduled job. it will not cause performance issue. Script will be executed using System scheduled job. once script execution is completed, the System scheduled job will be deleted automatically.

145) How to get the row count in a glide record?

Ans: By using the `getRowCount()` function you can retrieve the number of rows.

146) What is the difference between `deleteMultiple()` and `deleteRecord()`?

Ans: `deleteMultiple()` deletes multiple records according to the current "where" clause. Do not delete attachments, whereas `deleteRecord()` deletes the single record.

147) In which table update sets and customization are stored?

Ans: Each update set is stored in the Update Set [sys_update_set] table, and the customizations that are associated with the update set, are stored in [sys_update_xml] table.

148) Which method is used to get all the active/inactive records from a table?

Ans: You can use the addActiveQuery() method to get all the active records and the addInactiveQuery() to get all inactive records.

149) How do you get the result set from two tables in the glide script?

Ans: addJoinQuery(joinTable, [primaryField], [joinTableField])

Note: This is not a true DATABASE Join. addJoinQuery() adds a subQuery.

150) How do you get the records of specified fields which are not null?

Ans: addNotNullQuery(String fieldName) can be used.

Example: To get all the records where 'name' is not null.

```
addNotNullQuery('name');
```

151) How will you get all the records where the incident is having a category as hardware or software?

Ans: Use addOrCondition(String name, String oper, Object value) .

Example :

```
var gr = new GlideRecord('incident');  
var qc = gr.addQuery('category', 'hardware');  
qc.addOrCondition('category', 'software');  
gr.query();
```

152) How to determine whether any of the field values in a record has changed?

Ans: By using the method changes() you can determine that the field value has been changed for a record.

153) What is the difference between next() and _next() method?

Ans: The next() method is responsible to move to the next record in GlideRecord. _next() provides the same functionality as next(), intended to be used in cases when we query the table having a column name as next.

154) What is difference between initialise and newRecord method while inserting new record?

Answer)

initialize(): Creates record with empty default field values.

newRecord(): creates a GlideRecord, set the default values for the fields and assign a unique id to the record.

155) What is the use of get method in Glide record?

The 'get' method is used to return first record in the result set.

The 'get' method can also be used when you know the sys_id of that record. It returns Boolean values based on the result.

Example 1:

```
var grIncident=new GlideRecord('incident');
if(grIncident.get('9e7f9864532023004247ddeeff7b121f')){
gs.print('There exist incident with sys_id 9e7f9864532023004247ddeeff7b121f');
}else{ gs.print('There is no incident with sys_id 9e7f9864532023004247ddeeff7b121f');
}
```

Example 2:

```
var grIncident=new GlideRecord('incident');
if(grIncident.get('number','INC0000601')){
gs.print('There exist incident with number INC0000601');
}else{ gs.print('There is no incident with number INC0000601');
}
```

156)What is the necessity of script action if we already have script include to write server side script?

Answer)

Script Actions are executed asynchronously while Script Includes are executed synchronously.

If there is scenario where script execution is going to take longer time then it's always better to go with Script Actions as user don't need to wait for script execution completion.

157) How to stop form submission on Client and Server side?

Answer)

On client side, we can use 'return false' on onsubmit client script.

On server side, we use the function 'current.setAbortAction(true)' on before Business rule.

158) What are the private function in script include and how to define them?

Answer)

Function names starting with underscore are considered as private functions. These functions can be called only in same script include or extended/inherited script include.

159) How to copy attachments from one record to another?

Answer) `GlideSysAttachment.copy('from table', current.sys_id, 'to table', dmnd.sys_id);`

160) Which method of transferring update set is considered as best approach?

Answer: Transferring via update source.

161) Provide two issues that could be faced while previewing update set?

Answer)

1. Found a local update that is newer than this one.
2. Could not find a record in sys_scope for column sys_scope referenced in this update.

Note : There are many common preview errors we get so you can remember them if you come across it.

161) How can you fetch the last 5 incidents created?

Answer)

We can achieve it by using `orderByDesc()` and `setLimit()`.

```
var gr = new GlideRecord('incident');
gr.orderByDesc('sys_created_on');
gr.setLimit(5);
gr.query();
```

162) What is the use of `gs.include('validators')` ?

Answer:

Validators is basically a script include that allows you to validate the data coming in when you script is looking for input.

Things like isNumeric, or isInteger or containsOnlyChars

Example is on Priority check:

```
if (email.body.priority != undefined && isNumeric(email.body.priority))
```

isNumeric is checked in the script includes – validators – to make sure it's a number.

163) What is the use of sysparam_name in GlideAjax?

Answer: GlideAjax uses sysparm_name to find which function to use.

164) What do you mean by Function names starting with “_”?

Function names starting with “_” are considered private and are not callable from the client.

165) What is the use of gsftSubmit()?

gsftSubmit(null, g_form.getFormElement(), “UI Action Id”) triggers the UI Action which is specified in the 3rd parameter, which is the action name/element id. It is mostly used in UI actions that have a client side and a server side script.

At the end of the client side script, you call gsftSubmit in order to trigger the UI Action again – this time running only the server side code.

166) When Business Rule run?

Business rules run based on two sets of criteria:

The time that the business rule is configured to run relative to a record being modified or accessed.

Options	When the rule run
---------	-------------------

Before -	After the user submits the form but before any action is taken on the record in the database.
----------	---

After -	After the user submits the form and after any action is taken on the record in the database.
---------	--

Async -	When the scheduler runs the scheduled job created from the business rule. The system creates a scheduled job from the business rule after the user submits the form and after any action is taken on the record in the database.
---------	--

Display -	Before the form is presented to the user, just after the data is read from the database.
-----------	--

168) What are the different types of operation of ACL?

The different types of operations of ACL are listed below:

Execute: User cannot execute scripts on record or UI page.

Create: User cannot see the New UI action from forms. The user also cannot insert records into a table using API protocols such as web services.

Read: User cannot see the object in forms or lists. The user also cannot retrieve records using API protocols such as web services.

Write: User sees a read-only field in forms and lists, and the user cannot update records using API protocols such as web services.

Delete: User cannot see the Delete UI action from forms. The user also cannot remove records from a table using API protocols such as web services.

List_edit: User cannot update records (rows) from a list.

Report_on: User cannot create reports on the object.

Personalize_choices: User cannot right-click a choice list field and select Configure Choices

edit_task_relations: User cannot define relationships between task tables.

edit_ci_relations: User cannot define relationships between Configuration Item [cmdb_ci] tables.

save_as_template: Used to control the fields that should be saved when a template is created.

add_to_list: User cannot view or personalize specific columns in the list mechanic.

Note: A user must pass both field and table ACL rules in order to access a record object.

If a user fails a field ACL rule but passes a table ACL rule, the user is denied access to the field described by the field ACL rule.

If a user fails a table ACL rule, the user is denied access to all fields in the table even if the user previously passed a field ACL rule.

169) What is the objective of display BR?

The primary objective of display BR is to use a shared scratchpad object, `g_scratchpad`, which is also sent to the client as part of the form. This can be useful when you need to build client scripts that require server data that is not typically part of the record being displayed.

170) What is the difference between Actual elapsed time and Business elapsed time?

Actual elapsed values are calculated on a 24×7 basis.

Business elapsed values are calculated based on the schedule specified in the task SLA. The schedule is taken from the SLA definition by default.

171) What will be the output of Actual and Business Elapsed time of the SLA if there is no Schedule is attached?

If no schedule is specified, then the Business elapsed time is the same as the Actual elapsed time.

172) What do you mean by User Presence in Service Now?

User Presence allows you to see who is online when you are working in an instance.

Your avatar appears in the form header next to your name, and in multiple other places such as in activity streams, Visual Task Boards, live feeds, and Connect conversations. A dot on the avatar of the user represents online status.

> Green dot if the user is logged in.

> No dot if the user is not logged in.

> Orange dot if the user recently logged out.

By Default it is activated in every instance.

173) How can you disable the User Presence of your instance?

You can disable User Presence globally by changing the properties:

Navigate to sys_properties.list.

Locate the property named glide.ui.presence.disabled.

Set the Value to true.

174) What are the three parts of condition builder?

The three parts of condition builders are:

1. Field
2. Operator
3. Value

175) Define Activity streams.

An activity stream is a list of entries in records and conversations.

Examples of activity streams include journal fields like comments and work notes that display in task records and Connect Chat conversations.

176) What are the different types of Activity streams in Service Now?

There are three types of activity streams.

List activity stream

Record activity stream

Visual Task Boards

177) How can you add Tree Picker behavior to any field?

You can add tree picker by following the below steps:

Modify the dictionary entry for the field.

Add tree_picker=true to the Attributes field.

178) What is dot walking in Service Now?

Dot-walking provides access to fields on related tables from a form, list, or script.

If the current table contains a reference to another table, any field on the referenced table can be accessed using dot-walking. For example, `incident.assigned_to.company` references the company of the user assigned to an incident.

179) What is the use of User Preference in Service Now?

Individual users can configure many UI features. These user customization's are stored as records in the User Preference [`sys_user_preference`] table, and are updated each time the user changes the setting.

For example, by default the response time may appear at the bottom of lists and forms. If a user hides the response time, a user preference record is created showing the response time indicator as hidden. During the user's future sessions, the response time indicator is hidden. If the same user later displays the response time, the user preference record is updated appropriately and future sessions open with the response time indicator visible.

180) What do you mean by View in Service Now?

A view defines the elements that appear when a user opens a form or a list, and you can switch the view from the default for lists and forms.

181) Can we switch form views on a new form that has not been saved?

No, you cannot switch form views on a new form that has not been saved yet.

182) How can you redirect your banner logo link to any specific page? For example by default it redirects to home page, if you want to redirect to incident list page when we hit banner logo of the instance.

By the help of property (`glide.banner.image.url`) we can to control the banner logo URL. In the value field, change `home.do` to `incident_list.do`. It will redirect to incident list.

183) What is the use of the property `glide.banner.image.url_target`?

This property is used to redirect the browser windows when we click on the banner logo.

1. `gsft_main`: If you want to open in the same/content frame window.
2. `_top`: If you want to open in the full body of the window and replace the current window.
3. `_blank`: If you want to open in a new window/tab.

184) How can you add help to a field label on a form? For example:

1. Open the form.
2. Right-click the label for the field and select Configure Label.
3. Complete the form as per your requirements. You can also add URL.

185) How can you find out the all components (Tables, Modules, Script Include, Notifications, UI Action, etc.) installed with any application?

Follow the below steps:

1. In the navigation filter, enter: sys_metadata.list
2. Click the personalize list icon (gear icon), and add the Package column to the list.
3. Filter the list with these conditions to show only the components for the plugin:
4. [Package] [Is] [<Plugin name>]
5. Select Group by Class.

186) What is the use of ^ (carat)?

To build multiple term queries or specify multiple field values. For example:

sysparm_query=priority=1^incident_state=3

187) Define table in service now?

A table is a collection of records in the database.

Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table.

Table extension enable one or more child tables to share fields and records with a parent table.

188) Can we extend the table after creation?

No, you can only extend tables during table creation.

189) Define child class and parent class with respect to table administration?

A table that extends another table is called a child class, and the table it extends is the parent class.

A table can be both a parent and child class both extending and providing extensions for other tables.

190) What do you mean by base class with respect to table administration?

A parent class that is not an extension of another table is called a base class.

191) What are the different extension models present in now platform?

The Now Platform offers three extension models:

1. Table per class
2. Table per hierarchy
3. Table per partition

Table per class: Creates a separate database table for the parent class and each child class.

Table per hierarchy: Creates one database table for the parent class, which stores all records for the parent and child classes. Child classes do not have separate database tables.

Table per partition: Creates one database table for the parent class, which stores all records for the parent and child classes. Child classes do not have separate database tables. As the database table reaches a storage limit, the system dynamically adds storage tables (partitions) to store additional records.

192) How can you identify system defined table and custom tables?

System defined table mostly begins with sys and custom defined table always begins with u_ or x_ for scoped tables.

193) What are the global default fields added to all tables at the time of creation?

By default seven fields added to any table at the time of creation.

Field	Type	Description
Class [sys_class_name]	-	If the table is extensible, a string field that indicates which child table contains the record.
Created [sys_created_on]	-	A time-stamp field that indicates when a record was created.
Created by [sys_created_by]	-	A string field that indicates the user who created the record.
Sys_id [sys_id]	-	The unique record identifier for the record.
Updates [sys_mod_count]	-	A numeric field that counts the number of updates for this record since record creation.
Updated by [sys_updated_by]	-	A string field that indicates the user who most recently updated the record.
Updated [sys_updated_on]	-	A time-stamp field that indicates the date and time of the most recent update.

194) Which table stores the labels and language Information?

Field labels (sys_documentation) stores the labels and language information.

195) Can we delete base system tables?

Base system tables cannot be deleted. Also, any missing base system tables are recreated when the instance is upgraded.

196) What do you mean by Assignment rule?

Assignment Rules automatically assign tasks to users and groups.

Assignment rules do not apply to unsaved changes on a form. Also the rule is applied only if the task is not already assigned to another user or group.

197) What is difference between Assignment rule and Data lookup rule?

Data lookup rules offer a generic way to change any field value, not just assignment fields.

Data lookup allows rules to apply to unsaved changes on a form that is not possible using assignment rule.

You can override the default value using Data lookup but Assignment rules cannot overwrite existing assignments (including assignments set by a default value or a previously run assignment rule).

198) What is the difference between Additional Comments and work notes fields?

Additional comments: Any user can update.

Work notes: ITIL users can update.

199) How can you allow a field in a child table to have a different value or behavior than the same field in a parent table?

Answer: By the help of dictionary override.

200) Can you apply dictionary override to the tables present in different scope?

Answer: No, you can only add dictionary overrides on tables that are in the same scope as the parent table.

201) What do you understand by "Collection" field type in service now?

Dictionary entries with the Collection type represent the table rather than a field on the table. Changes such as attributes or the Read only check box applied to this entry are applied to the table rather than a specific field. This entry is automatically created when a table is created. There should not be more than one entry for a table of this type.

202) How to move the development which is not captured in update set?

Answer) By exporting the xml manually and importing in the target instance.

203) Where are the additional comments are stored in ServiceNow Database?

Answer) Table name is 'sys_journal_field'

204) Where are history stored in ServiceNow?

Answer) History activity are captured in various tables like:

sys_history_set

sys_history_line

205) How to make filter dynamic on a field based on other field in serviceNow?

Answer) var collectorName = 'cn';

var filterString = "";

var dynamicQuery = "" ;

var order=g_form.getValue('autopopulatedvariable');

dynamicQuery = "u_order="+order;

try{

var myListCollector = g_list.get(collectorName);

myListCollector.reset();

myListCollector.setQuery(dynamicQuery);

}

catch(e){

//Reset the filter query

window[collectorName + 'g_filter'].reset();

window[collectorName + 'g_filter'].setQuery(dynamicQuery);

window[collectorName + 'acRequest'](null);

}

206) How to query oldest 5 created record on table?

Answer)

You can order the results of your recordset by using 'orderBy' and/or 'orderByDesc' as shown below.

//Find all active incidents and order the results ascending by category then descending by created date

var gr = new GlideRecord('incident');

gr.addQuery('active', true);

gr.orderBy('category');

gr.orderByDesc('sys_created_on');

gr.query();

207) What is GlideAggregate in ServiceNow?

Answer) GlideAggregate is actually an extension of the GlideRecord object. It allows you to perform the following aggregations on query recordsets...

- COUNT
- SUM
- MIN
- MAX
- AVG

```
//Find all active incidents and log a count of records to the system log
var gr = new GlideAggregate('incident');
gr.addQuery('active', true);
gr.addAggregate('COUNT');
gr.query();
var incidents = 0;
if (gr.next()){
incidents = gr.getAggregate('COUNT');
gs.log('Active incident count: ' + incidents);
}
```

208) What is encodedQuery in ServiceNow?

Answer) My encoded query string would be this...

active=true^category=software^ORcategory=hardware

I could build that encoded query string and use it in a query like this...

```
//Find all active incidents where the category is software or hardware
var gr = new GlideRecord('incident');
var strQuery = 'active=true';
strQuery = strQuery + '^category=software';
strQuery = strQuery + '^ORcategory=hardware';
gr.addEncodedQuery(strQuery);
gr.query();
```

209) How to get Row Count on table in ServiceNow?

Answer) Using getRowCount() and GlideAggregate, But using GlideAggregate is good practice.

210) How to identify duplicate records?

Answer)

```
var gr = new GlideAggregate('incident');
gr.addAggregate('COUNT', 'number');
gr.addHaving('COUNT', '>', 1);
gr.groupBy('number');
gr.query();
while (gr.next()) {
    var number = gr.getValue('number');
    var count = gr.getAggregate('COUNT', 'number');
    var duplicateRecords = new GlideRecord('incident');
    duplicateRecords.addQuery('number', number);
    duplicateRecords.query();
    while (duplicateRecords.next()) {
        gs.print('Incident Sys ID: ' + duplicateRecords.getValue('sys_id') + ', Created On: ' + duplicateRecords.getValue('sys_created_on'));
    }
}
```

211) What is getRefRecord()?

Answer) The 'getRefRecord' method can be used as a shortcut to query a record populated in a reference field on a record.

```
var caller = current.caller_id.getRefRecord(); //Returns the GlideRecord for the value populated in the 'caller_id' field
caller.email = 'test@test.com';
caller.update();
```

212) How to create a record using Script?

Answer) //Create a new Incident record and populate the fields with the values below

Inserts are performed in the same way as queries except you need to replace the 'query()' line with an 'initialize()' line as shown here.

```
var gr = new GlideRecord('incident');
gr.initialize();
gr.short_description = 'Network problem';
gr.category = 'software';
gr.caller_id.setDisplayValue('Joe Employee');
gr.insert();
```

213) How to update a record using script?

Answer) You can perform updates on one or many records simply by querying the records, setting the appropriate values on those records, and calling 'update()' for each record.

```
//Find all active incident records and make them inactive
var gr = new GlideRecord('incident');
gr.addQuery('active',true);
gr.query();
while (gr.next()) {
gr.active = false;
gr.update();
}
```

214) How to delete Records using script?

Answer)

Delete records by performing a glideRecord query and then using the 'deleteRecord' method.

```
//Find all inactive incident records and delete them one-by-one
var gr = new GlideRecord('incident');
gr.addQuery('active',false);
gr.query();
while (gr.next()) {
//Delete each record in the query result set
gr.deleteRecord();
}
```

215) Write a script to show incident count for each state which are updated today?

We can use GlideAggregate API to achieve this scenario:

Script :

```
gs.print('State\\tIncident Count');
var grIncident=new GlideAggregate('incident');
```

```

grIncident.addEncodedQuery(' sys_updated_onONToday@javascript: gs.beginningOfToday()@javascript: gs.endOfToday()');
grIncident.addAggregate('COUNT','state');
grIncident.query();
while(grIncident.next()){
gs.print(grIncident.state.getDisplayValue()+'\t'+grIncident.getAggregate('COUNT','state'));
}

```

216) Write a script to show incident count for each category?

We can use GlideAggregate API to achieve this scenario:

Script :

```

gs.print('State\\tIncident Count');
var grIncident=new GlideAggregate('incident');
grIncident.addAggregate('COUNT','category');
grIncident.query();
while(grIncident.next()){
gs.print(grIncident.category.getDisplayValue()+'\t'+grIncident.getAggregate('COUNT','category'));
}

```

217) Write a script to show incident count for each priority?

We can use GlideAggregate API to achieve this scenario:

Script :

```

gs.print('State\\tIncident Count');
var grIncident=new GlideAggregate('incident');
grIncident.addAggregate('COUNT','priority');
grIncident.query();
while(grIncident.next()){
gs.print(grIncident.priority.getDisplayValue()+'\t'+grIncident.getAggregate('COUNT','priority'));
}

```

219) How to delete duplicate records?

Answer:

```

var testGr = new GlideAggregate('u_test_data'); // table on which deletion is to be performed

```



```

testGr.groupBy('u_test_data_column2'); // column which you think has duplicate values i think it would be the value column for sys_choice table
testGr.query();
var testGR1;
while (testGr.next()) {
    testGR1 = new GlideRecord('u_test_data');
    testGR1.addQuery('u_test_data_column2', testGr.u_test_data_column2);
    testGR1.query();
    testGR1.next(); // Skip the first result
    while (testGR1.next()) { // delete the next one
        testGR1.deleteRecord();
    }
}

```

220) How to get instance name using script?

Answer) Using system property, gs.getProperty('instance_name');

221) What is the difference between .getValue() and .getDisplayValue()?

Answer) .getValue() will give you only backend value(sys id, choice value), .getDisplayValue() will give you display name of the value(display name).

222) Does Scheduled Jobs get captured in update sets:

Answer) By default, Scheduled jobs are not captured in update sets.

223) Can we capture The development in update sets which are not captured in update sets?

Answer)

- 1) Using Script, we can capture in updatesets.
- 2) Using Export XML in source, Import XML in target instance.

224) How to call Script Include in Server Side?

Answer) new scriptInclude().methodName();
 Note: you can pass arguments at script include level
 new scriptInclude(current).methodName();
 You can arguments at method Level.
 new scriptInclude().methodName(current);

225) How to call Ui Pages in ServiceNow?

Using GlideDialogWindow or GlideModal, we call Ui pages in ServiceNow

Syntax:

```
var gwPortals = new GlideDialogWindow('try_item_portal_page');
gwPortals.setTitle('Try Portal Page (select portal)');
gwPortals.setPreference('portal_page_id', 'sc_cat_item');
gwPortals.setPreference('item_id', itemID);
gwPortals.render();
```

226) How to pass values from Client Side(UI Action, Client Script, UI Policy, UI Macro) to UI Pages in ServiceNow?

Answer: using setPreference method in GlideModal.

```
gwPortals.setPreference('portal_page_id', 'sc_cat_item');
gwPortals.setPreference('item_id', itemID);
```

227) How to access values in Ui pages from client side?

Answer: using below line, we can access values from client side UI page.

```
<g:evaluate var="jvar_text" expression ="RP.getWindowProperties().get('text')"> </g:evaluate>
```

228) How to close UI page using script in Servicenow?

Answer:

```
GlideDialogWindow.get().destroy();
```

229) How do you access HTML values in Client Side Script in UI Pages?

Answer: using gel. gel is like document.getElementById('ID name');

```
var assiTo = gel('ID name').value;
```

230) What is processing Script in ServiceNow?

Answer: Processing script is like server side script where you can write javascript, Server Side APIs
This will be executed when we write HTML within <g:ui_form></g:ui_form> tags.

231) How to call table form directly using UI Actions:

Answer: Using GlideModalForm, we can directly call the table form using script.

232) How many places we write conditions in Business Rules?

Answer: Condition Builder

Condition

Roles

Inside the script

233) how to redirect to same URL in UI actions?

Answer: `action.setRedirectURL(current);`

234) How to override UI actions on child table?

Answer: whenever a UI action is created it will be shown on child tables to avoid the we have to use condition using script

`current.sys_class_name == 'incident'`

235) How to get current date?

Answer:

`new GlideDateTime(gs.nowDateTime())`

236) How to identify today is weekday or weekend?

Answer: Using `GlideDateTime()`

`var today = new GlideDateTime();`

`var day = today.getDayOfTheWeek();`

`if(day != 6 || day != 7){`

```
gs.print('Weekday');  
}else{  
gs.print('Weekend');  
}
```

237) How to get date difference between two dates?

Answer:

```
gs.dateDiff('2024-23-08 00:00:00', '2024-30-08 00:00:00', true);
```

true will give you output in seconds

```
gs.dateDiff('2024-23-08 00:00:00', '2024-30-08 00:00:00', false);
```

false will give you output in days.

```
GlideDateTime.subtract('2024-23-08 00:00:00', '2024-30-08 00:00:00').getDayPart();
```

238) Types of Notifications in ServiceNow?

Answer:

Email
Push
Provider

239) How many ways we can send email notifications from ServiceNow?

Answer:

When record inserted or updated
When event fired
Triggered(Flow Designer)
Workflow.

240) How to send email using script:

Answer: `gs.eventQueue('event_name', obj, 'eventparm1', 'eventparm2');`

241) Where do we find if the event is triggered in Servicenow?

Answer: We will find in 'event log' under application navigater.

242) How to call email script in Email Notifications?

Answer: \${mail_script:emailsriptName}

243) How to show record URL in Email Notification in ServiceNow?

Answer: \${URI}

244) How to show record URL with number in Email Notification in ServiceNow?

Answer: \${URI_REF}

245) Some of the Server side best practices in Business Rules?

Answer:

- Dont Using 'gr' variable for every gliderecord
- Dont using current.update() in server side script
- Dont Using before business rules for creating or updating records.
- Dont using synchronous calls
- Dont write conditions within the script field

246) Some of the best practices in Server side?

Answer:

- Dont Using 'gr' variable for every gliderecord
- Dont using current.update() in server side script
- Dont Using before business rules for creating or updating records.
- Dont using synchronous calls
- Dont write conditions within the script field
- Removing the system logs in script after debugging
- Dont use repetative script, use functions instead.

Using Flow Designer instead of writing script, servicenow prefers low code.
Adding Roles to the Group instead of adding to the user account.
Commenting your code to make every developer understand
Using GlideAggregate for aggregations such as count, min, max, avg
dont use gerRowCount for aggregations
Using GlideRecordSecure is good practice.
using setLimit when updating or deleting
Use Descriptive Variable and Function Names
Use Variables to Store Function Results
Verify Values Exist Before Using Them
Return Values
Avoid Dynamic JEXL Expressions in an Evaluate
Avoid the eval Function
Using System Properties instead of hardcoded values
Using Messages, instead of writing messages in the `gs.addInfoMessage()`, `gs.addErrorMessage()`;

Please follow below document for future reference

https://developer.servicenow.com/dev.do#!/guides/washingtondc/now-platform/tpb-guide/logs_and_queues_best_practices

247) How to set subject dynamic in Email Notifications?

Answer: `email.setSubject('subject here');`

248) How to set email body dynamic?

Answer: `email.setBody('body here');`

249) How to keep users in copied list in ServiceNow?

Answer: `email.addAddress('cc','email/sysid', 'DisplayName');`

250) How to receive event values in Email scripts?

Answer: using event object, event object has access to event log record that is created when event is fired.

251) What is email_action in email script?

Answer: it will hold email notification record values in email script, such as email name, category, recipients..etcc

252) How to create related list?

Answer: To create new related list, we have to navigate to relation ships under system definition.

current object refers to Querys to table

parent object refers to applies to table

253) Where does the attachments get stored in serviceNow?

Answer: sys_attachment table

254) How to create a new attachment using script in servicenow?

Answer: try using the below code snippet ,I hope this will help you.

```
var attach=new Attachment();  
  
attach.write(tablename,rec sysid,filename,content type,value) ;  
  
//here tablename is the table were you need to attach the attachment.  
// rec sysID for the record on which you need to attach the attachment.  
// File name is the name of attachment file with extension like "filename.pdf".  
//content type is the extension for file in your case it is "application/pdf".  
//value is the base64 of the attachment file .
```

255) What are update Set Best practices?

Answer:

Never Delete Updates, Instead Move Them to Default

Check All Updates Before Migrating, Remove Unnecessary Updates,Once Again, Keep Update Sets Small

Publish Workflows Before Completing an Update Set

Update Set Naming Convention

Don't forget to add story number in update set.

Specify Update Set Order

Don't Merge Update Sets

Keep the Number of Updates per Update Set Small

Update Set Migration Order

Promote Changes After Hours or on Weekends

256) how to move updates from one update set to another update set?

Answer:

Complete the update set in source instance and download the XML file. Import the xml file on target instance, Preview the update set and commit the update set.

Using Update source, Retrieve the update set using update source in target instance, preview update set and commit update set.

257) What is Global Business Rule?

Answer: A global Business Rule is any Business Rule where the selected Table is Global. Any other script can call global Business Rules. Global Business Rules have no condition or table restrictions and load on every page in the system.

258) What is Query Business Rule?

Answer: Query business rules will be executed when the data loaded when you open the list or form. Query business rules let you run code and possibly modify a query before/after the query is executed. This is useful if you don't want anyone to see inactive users. There's a OOB business rule doing this for non-admins.

Display business rules run when the data is loaded.

259) what is on demand scheduled job?

Answer: This is for manual execution. After you save it, you click "Execute Now" in the top right and it runs. It will not run automatically.

260) Can we rollback when we run the background script?

Answer: Yes you can. You can roll back the database actions of a script executed using the Scripts-Background module. When you executed the script using the Scripts-Background module, the Record for Rollback? item must have been checked.

You can go to script execution history and click on rollback.

261) what is Display BR and explain with examples?

Answer: It will run only when the form gets loaded, But the Display BR will run in the server side. We can only use g_scratchpad in Display Business rules. We can pass values from Display Business Rules to client side using g_scratchpad object.

262) when display BR will run?

Answer: Display business rules will run before the client scripts and UI policies

263) What is the script execution order in servicenow?

Answer:

Query Business Rules
Display Business Rules
onload Client Script
onload UI policy
onchange Client script
onchange UI policy
onsubmit client script
Client UI Actions

Server Side UI Actions

Before business rules: Scripts configured to execute before the database operation with an order less than 1000.

Before engines. The following are not executed in any specific order:

Approval engine (for task and sys_approval_approver tables)

Assignment rules engine (for task tables)

Data policy engine

Escalation engine

Field normalization engine

Role engine - keeps role changes in sync with sys_user_has_role table (for sys_user, sys_user_group, sys_user_grmember, and sys_user_role tables)

Execution plan engine (for task tables)

Update version engine - creates version entry when sys_update_xml entry is written (for sys_update_xml table)

Workflow engine (for default workflows)

Before business rules: Scripts configured to execute before the database operation with an order greater than or equal to 1000.

The data base operation (insert, update, delete).

After business rules: Scripts configured to execute after the database operation with an order less than 1000.

After engines. The following are not executed in any specific order:

Label engine

Listener engine

Table notifications engine

Role engine - keeps role changes in sync with sys_user_has_role table (for sys_user, sys_user_group, sys_user_grmember and sys_user_role tables)

Text indexing engine

Update sync engine

Data lookup engine inserts or updates

Workflow engine (for deferred workflows)

Trigger engine (for all Flow Designer flows)

Email notifications. The following are executed based on the weight of the notification record:

Notifications sent on an insert, update, or delete

Event-based notifications

After business rules. Scripts configured to execute after the database operation with an order greater than or equal to 1000.

264) what is g_scratchpad and how to use it , explain with example?

Answer: We can only use g_scratchpad in Display Business rules. We can pass values from Display Business Rules to client side using g_scratchpad object.

265) What is script include, Explain with examples?

Answer: The Script Include function is defined using standard JavaScript syntax. On demand Script Includes are typically used when script logic needs to be reused. Examples include standardizing date formats, enabling/disabling logging, and validating email addresses.

3types of script Includes:

Class (Script Include with function)

Class less(Normal Script Include)

Extended Class(Client Callabale Script Include)

266) Difference between script include vs BR?

Answer:

Script includes run only when called by a server script.

Business rule: A business rule is a server-side script that runs when a record is displayed, inserted, updated, deleted, or when a table is queried.

267) how to call script include in the Server Side?

Answer: new scriptIncludeName().methodName();

268) what is dynamic reference qualifier and explain with examples?

Answer: It dynamically changes references or qualifiers based on user input into the fields of the current form.

When you create a dynamic Reference Qualifier on Reference field referencing user table, You can use it for all the fields which is reference and referencing user table.

Active User: 'active=true'

269) what is data policy and explain with examples?

Answer: Data policies can apply rules to all data entered into the system, including data brought in through import sets or web services and data entered through the mobile UI. For example, suppose that you are configuring a web service that allows users from outside the platform to update problems on the ServiceNow instance.

When you are trying to resolve incident from different sources, resolution code and notes are mandatory from all the sources, then we will use data policy.

270) what are the rule need to follow to convert ui policy to data policy?

Answer: When u r not controlling the data visibility, When you are not controlling the views, When you are not controlling the related lists, When you are not controlling the script.

271) Can we use 'previous' object in Async business rule?

Answer: No, We cannot , Previous object is not available in Async business Rules.

272) What is the difference between after and Async Business Rule?

Answer: After business Rules run synchronously, Async business Rules run asynchronously. After BRs will wait to the complete script execution and it will cause performance issue. Async BRs will run asynchronously and it will run in the background using a system scheduled job. it will not cause performance issue. Script will be executed using System scheduled job. once script execution is completed, the System scheduled job will be deleted automatically.

273) What is the difference between Before and After Business Rule?

Answer:

Before Business Rules run before data gets stored in the database.

After/Async business Rules run after data gets stored in the database.

274) How many types of business rules in ServiceNow and Explain about each business rule?

Answer: 4 Types of Business Rules. Before, After, Async, Display.

Before Business Rules run before data gets stored in the database.

After/Async business Rules run after data gets stored in the database.

Display Business Rules run everytime when the form gets loaded.

275) what is service catalogs and explain with example?

Answer: Service Catalog enables to organize items into multiple categories and sub- categories to provide access and use by all users. The best practice is to create these categories in a user-friendly way while avoiding complex category trees.

276) what is self-service?

Answer:"Self service" refers to a system or a process where users can perform tasks, obtain information, or complete transactions independently without requiring assistance from a third party, such as a customer service representative or a support agent. The concept is commonly used in various industries and contexts, including IT services, retail, banking, and customer support.

278) what are the ticket(record) creation channels?

Answer: Email, Script , API, Form, Virtual Agent, Record Producer, Self Service...Etc

279) what is the latest catalog item created by you and explain the flow and purpose?

Answer: Please refer Youtube for detailed catalog item creation.

280) what is category in catalog Items?

Answer: Category is to separate the Catalog Items as a group.

281)what is catalogs option in cat item page?

Answer: Catalogs allow grouping of catalog items based on target user base. A catalog is where Items are held (Maintain Items).

282) what is variables and explain few important types?

Answer: variables are like fields in the catalog items, Ex. Single line text, Multi Line Text, Reference, Select Box, Container Start, End, Split, Multiple Choice.

283)what is variable sets?

Answer: Using variable sets saves time because you do not have to create the same variables individually for many catalog items. Also, when variables should be modified, you can modify the variable set and the changes are reflected across all the catalog items that are associated with the variable set.

284)what is the main use of variable sets, explain with example?

Answer:Using variable sets saves time because you do not have to create the same variables individually for many catalog items. Also, when variables should be modified, you can modify the variable set and the changes are reflected across all the catalog items that are associated with the variable set.

285)what is single row and multi row variable set?

Answer: Use a single-row variable set to capture data from variables that are grouped together, while submitting a catalog item request for a single entity. Multi-row variable set. Use a multi-row variable set to capture variable data in a grid layout while submitting a catalog item request for a group of entities.

286)how you read values of multi row variable set?

Answer:

```
var mrvsRows = g_form.getValue("mrsv_name");  
//The above will return a string in the form of array and objects. Now you will have to parse it
```

```
var oMrvs = JSON.parse(mrvsRows);  
var sum = 0;  
for(var i in oMrvs) {  
    sum += oMrvs[i]["cost_field_name"]  
}
```

```
g_form.setValue("total_cost_field_name", sum);
```

287) if we create a ui policy or client script on variable set level OR catalog item level - what is the difference?

Answer:

Catalog Item client script will run first and Variable set catalog client script will run after that.
Catalog Item UI Policy will run first and Variable set catalog UI Policy will run after that.

288) what is the catalog ui policy and difference between normal ui policy to cat ui policy?

Answer: Catalog Item UI Policy will not control views and it will work on catalog form, requested item form, catalog task form.

289) what is the catalog client script and difference between normal client script to cat client script?

Answer:

There are 4 types of client scripts in Normal Client scripts

There are 3 types of client scripts in Catalog Client Scripts

Catalog Client scripts work on Catalog Items, Requested Items, Catalog Tasks.

Normal Client Scripts work on views, Catalog Client Scripts will not work on views.

290) How to make Ui Policy work only work on Record Producer?

Answer: We should only check Applies on target Record check box on Catalog UI Policy.

291) How to make Ui Policy work only work on Catalog Tasks?

Answer: We should only check Applies on Catalog Check Box check box in catalog Client Script in catalog items.

292) what is user criteria?

Answer: User Criteria is used to restrict the access to catalog items, Record Producers, Knowledge Articles.

293) how to limit the access of a catalog items?

Answer: In the available for, we have to create or add User Criteria.

294) what is available for or not available for?

Answer: Both available for, not available for, uses User Criteria to control the visibility of the catalog item.

Available For means who can see catalog items

Not available for means Who cannot see catalog items

295) how many catalog items you created & explain few?

Answer: I created 25 catalog Items in my in development journey.

296) can we attach two categories for one catalog item?

Answer: Yes, you can, In the catalog Item related list, we can add multiple category's for one catalog.

297) what is the difference between available for vs not available for?

Answer: Bot available for, not available for, uses User Criteria to control the visibility of the catalog item.

Available For means who can see catalog items

Not available for means Who cannot see catalog items

298) where are the places "user criteria" will be used.?

Answer: User Criteria is used to restrict the catalog items access and Knowledge Articles access.

299) what is record producer, explain with Example?

Answer: Record producers use the Catalog Item user interface to create records in application-specific tables. such as Incident, problem, change Request, User, Group ..Etc

When Record Producer form is submitted, It will create a record on target table.

300) what is the difference between catalog item vs record producer?

Answer: If you're offering end-users the ability to request a service where you need to control approvals and fulfillment, you will use a catalog item. If you're just looking to gather information or need to gather open-ended records from your users, you will use a record producer. Record producer is used to create records in servicenow tables.

301) can we attach a workflow to a record producer?

Answer: We cannot directly attach workflow to Record Producers, We can attach workflow to target record

302) can we trigger a workflow from record producer?

Answer: Yes, We can trigger workflow from Record Producer using script
`new Workflow().startFlow('wf_sys_id', current, current.operation())`

303) how to read variable value from record producer script?

Answer: Using Producer object

producer.requested_for
producer.location

304) what are the way you can map variable value to field value of the record which is going to be generated using record producer?

Answer: Field Map, Script in record producer using producer object.

305) how many record producers you created so far in your career?

Answer: Refer the youtube videos for any record producers.

306) what is order guide in servicenow?

Answer:
Typically an order guide is used to group catalog items together so that they can be ordered at the same time (ex a "New Hire" order guide that gives someone the option to order a laptop, dock, monitors, etc). Each catalog item would have its own flow to handle how it should be fulfilled.

307) what is the difference between order guide vs catalog item?

- Answer:
1. Catalog/Maintain items is used to create and maintain individual catalog items - like iPad, Laptop, etc. where as Order guides are used to create a "collection" of related items - like a new employee. You can put your related items (created from Maintain Items) into one cohesive collection with options to select those items and fill in the details.
 2. The main difference between a catalog item and an order guide is what happens after you submit them. A catalog item works with the cart where you can add multiple and then checkout. On the back end it creates a request, request item, and possibly approvals and tasks depending on its workflow. A record producer is ordered one at a time and creates one record on the back end, most commonly an incident. An order guide is used to guide the user into ordering multiple items.
 3. A Catalog item request has only a single request item under it. where as an order guide can have multiple RITMs under it based on Rule Base.

4. Catalog Items and Record Producers are forms that people can fill out that will create records after being submitted whereas Order guides are a way to bundle catalog items together. Variables are the questions/fields that are asked on the catalog items, record producers, and order guides. They can be associated with those forms.

5. Order guides can handle multiple requests at the requested item level and still present it as one single item to the user. Behind the scenes you will create either a record producer or a catalog item for each access form. And then include these forms in the order guide. You can decide which request form to show based on user input on a previous page. By this I mean, the user will answer a few questions (you will put them as variables on the order guide) to describe the need. And based on those answers, you will show the access forms. When user submits the order guide, it will create multiple requests for each form

308) Can you write client APIs in record producer script ?

Answer: No, We can't write client side APIs in Record Producer Server Side Script.

309) what is cascade variable?

Answer:

The cascade variable checkbox in an order guide in ServiceNow allows for common variables to be updated automatically across all items in an order guide

Variable name in order guide should match with Variable name in catalog item to pass values.

310) what is rule base and explain with example?

Answer: order guide uses the rule base that means add the catalog item based on some rule. The purpose of a rule base on an order guide includes catalog items into your order depending on when you want to include the item.

311) how many variable sets you created in your career?

Answer: I created 10 variable sets.

312) what is workflow?

Answer:

Part of the ServiceNow platform that automates the multi-step process. A Workflow is defined as a series of Activities that follow execution paths. An Activity is a set of instructions that can include script, logging, approvals, record processing, timers, and wait time.

313) what is checkout and publish?

Answer:

To edit the workflow, we have to check out.

Once you completed the workflow, You have to publish the workflow.

314) what is begin and end?

Answer: When workflow is created begin and End activities will be created by default.

315) in which table, workflow will run as per request fulfillment process?

Answer: wf_workflow_context

316) what are the activities you know?

Answer:

approval user -- To send approvals to individual user accounts

approval group -- To send approvals to individual user group

approval action -- To set approvals status on Requested Item

approval coordinator -- To send User, Group approvals at a time.

if -- This activity is used to check conditions.

switch -- This activity is used to split conditions.

wait for condition -- for advanced conditions, if we have to wait for manual and automated tasks.

timer -- to hold the workflow for certain time.

317) Can you please explain the complex workflow you created?

Answer: Refer youtube videos.

318) Explain below activities?

approval user -- To send approvals to individual user accounts

approval group -- To send approvals to individual user group

approval action -- To set approvals status on Requested Item

approval coordinator -- To send User, Group approvals at a time.

if -- This activity is used to check conditions.

switch -- This activity is used to split conditions.

wait for condition -- for advanced conditions, if we have to wait for manual and automated tasks.

timer -- to hold the workflow for certain time.

319) what is the difference between "create task" & catalog task?

Answer:

Create task will create record on task based table such as Incident, Problem, Change Request, Incident Task, Problem Task....

Catalog Task will create records on sc_task table.

320)how to create notification from workflow?

Answer: Using Notification activity in workflow, Create Event activity.

321)what is parallel activities and what is the known issue with parallel activity?

Answer: When we use activity's parallelly instead of Sequential activitys, the activitys doesnot wait.

322) what is branch and join?

Answer: the Branch workflow activity behaves exactly the same as if you were to draw multiple transition lines from a single exit condition, whereas the Join workflow activity allows you to merge multiple transitions into a single path.

If we create multiple tasks at a time parallelly, It has to wait for all the tasks to be completed, Then we will use join activity.

323) what is turnstile and rollback activitys in servicenow?

Answer:

The Turnstile activity limits how many times a workflow can pass through the same point.

Rollback to activity will move the workflow towards backward direction.

324)what is run script activity?

Answer: Run Script activity is used to write any server side script in workflow.

325)What are the workflows that you created so far?

Answer: Please go through the videos.

326) what is the difference between workflow vs flow designer?

Answer:

Flow Designer is used as Business Rules

Flow Designer is used as Scheduled Jobs

Flow Designer is used as Emails

Flow Designer is used as Inbound Email Notifications

Flow Designer is used as Workflow

Flow Designer is used as Integration Hub

327) what is subflow?

Answer: Subflow in flow designer is set of actions with inputs and outputs.

328)how to call a value from activity to activity?

Answer: workflow.scratchpad,variable_name

329) how to pass a value from mainworkflow to subflow?

Answer: We have to create inputs in Child workflow. We can pass values from parent workflow to child workflow, In the child workflow, we can access them using workflow.inputs.variable_name.

330)what is stages?

Answer:workflows are structured processes that automate tasks and activities. They consist of various stages that help in visualizing and managing the process flow.

Wait for approval

Request Approved

Fulfillment

Delivery

Request Cancelled

Completed

331) what is workflow_scratchpad?

Answer: To carry values from one activity to another activity in workflow, we use workflow.scratchpad.variable_name

332) How to trigger workflow using script?

Answer:

```
var workflow = new Workflow();  
// Start the workflow on the current record  
workflow.startFlow('your_workflow_sys_id_or_name', current);
```

333)How to cancel the workflow using script?

Answer:

```
var workflow = new Workflow();  
  
// Query for the workflow context  
var wfContext = new GlideRecord('wf_context');  
wfContext.addQuery('table', current.getTableName());  
wfContext.addQuery('id', current.sys_id);  
wfContext.addActiveQuery();  
wfContext.query();  
  
while (wfContext.next()) {  
    // Cancel each workflow context found  
    workflow.cancel(wfContext);  
}
```

334)How to restart the workflow using script?

Answer:

```
var workflowUtils = new Workflow();  
workflowUtils.restartWorkflow(current, 'your_workflow_sys_id_or_name');
```

335) How to fire workflow event using script?

Answer:

```
var wf = new Workflow().getRunningFlows(reqItem);
```

```
while(wf.next()) {  
  new Workflow().broadcastEvent(wf.sys_id, "ENTER YOUR EVENT NAME HERE");  
}
```

336) what kind of issues you faced from workflow?

Answer: Unable to pass values from main workflow to child workflow. Because We cant create inputs on child workflow which is on sc_req_item table. I changed the table name and created inputs and changed the table back to sc_req_item.

337)what are workflow versions?

Answer: Everytime when you publish the workflow, a new workflow version will be created.

338) what are the main tables of workflows?

Answer:wf_workflow, wf_workflow_context, wf_workflow_version,wf_history,wf_activity,wf_transition

339)what are the tables that stores information about service catalog concepts?

Answer:

Catalog Variables (item_option_new): Stores definitions for variables used in catalog items

Catalog Items table: sc_cat_item

sc_request

sc_req_item

sc_task

sc_item_option

sc_item_option_mtom

sc_multi_row_question_answer

340) what is the meaning of "self service"?

Answer:

"Self service" refers to a system or a process where users can perform tasks, obtain information, or complete transactions independently without requiring assistance from a third party, such as a customer service representative or a support agent. The concept is commonly used in various industries and contexts, including IT services, retail, banking, and customer support.

341) what are the channels to create a ticket?

Answer:

Service Portal
Email
Excel
Data Source
Script
External Sources

342) what is the integration and explain with examples?

Answer: ServiceNow application can integrate with external systems such as Salesforce, AWS, AZURE, Power BI.
Ex. When record is created in ServiceNow, Record has to be created in external systems.

343) what is data source and what are the types of data sources available, explain with examples?

Answer:
LDAD
EXCEL
JDBC
OIDC
FILE

344) explain the process of ETL process in ServiceNow?

Answer:

Extract, Transform, Load

The ETL process in ServiceNow facilitates seamless integration of data from external sources into the platform's ecosystem, supporting various functionalities such as IT service management, asset management, and reporting. By effectively extracting, transforming, and loading data, organizations can centralize information in ServiceNow, enhance operational efficiency, and derive actionable insights from consolidated data sources.

345) what is import set?

Answer: When we load the data from different sources like Excel, JDBC...Etc, Import Set will be created.

346) what is transform map?

Answer: Transform Maps are configured and managed through the ServiceNow UI, typically within the Import Sets and Transform Maps module. Administrators and developers define and refine Transform Maps based on specific integration requirements, ensuring seamless data synchronization and integration between external systems and ServiceNow.

When we load the data in ServiceNow, The data will be loaded in temporary table, To transform loaded data from temporary table to target table in servicenow, we use transform map.

347) Explain transform scripts and types with examples?

Answer:

onStart: onStart runs at the very beginning of the import before any of the records are processed. None of the fields can be accessed on the source or target tables since they have not been set up yet. If you try to access fields in the onStart script, the transform will not run. If you wanted to inform a user that more records were being imported, you could use the onStart script to send out the message that the import has started.

onBefore: onBefore runs before every row is processed. This script runs for each individual record. If you want to set or alter values on the source table (see Temperature example explained in the Field Level Source Script section below) it can be done here. onBefore scripts can be used to verify that the value of a source field is valid. If the value is invalid, the script can take action.

sourceScript: The source script replaces the source field in the field map. It gives the target field a value. This is used when you want to change an individual field on each record. For example, if I had a Temperature field in my source table that was in Fahrenheit and I wanted it to be in Celsius on my target table, I would do the conversion in the field level source script. This will modify the temperature value for every record.

onForeignInsert: onForeignInsert runs before a new referenced record is created. Since this script only runs on an insert, the choice action “create” has to be selected for that field map. If “create” is not selected, the script will never run.

onChoiceCreate: onChoiceCreate runs before a new choice is created on a choice field. Since this script runs when a choice is created, “create” has to be selected as the choice action for that field map. If it is not selected, the script will not run. Because onChoiceCreate is a before action, only the source table record has set values. Similar to onForeignInsert, the onChoiceCreate script can take additional action when a new choice is created on a field.

onReject: onReject runs before a record (often a referenced record) is rejected. In order for this script to run, “reject” has to be selected as the choice action for that field map.

Record Level Script:

Record level scripts run alongside the field maps as they transform fields from the source table to the target table. Both the target and source tables are accessible for this script. Since the script will execute for every record, the record level script is where you want to alter record values. If I had a Firstname and Lastname field on my source table and I wanted to merge them into a Name field on my target table, I would do that with the record level script since I have access to all the fields in both the source and target tables.

onAfter: onAfter runs after every row has been processed. Fields on the target table can be accessed from this point. They contain the value that was just inserted. At the time that this script runs, the source and target field values should be the same. With onAfter scripts, you can take the values of the fields that were just inputted and use them to fill out record fields on other tables.

onComplete: onComplete runs once the import has been completed and all records have been transformed. The script will only run once. As an example, you could use the onComplete script to send out a message saying that new records have just been added to the table.

VARIABLES

There are many variables that can be used in transform scripts. Here is a list of all the variables and how they work:

action – returns “insert” or “update” depending on if the current record was or is being added or updated

error – a boolean that will stop the transformation and send an error message if set to true

error_message – the message sent if an error occurs

ignore – a boolean that will stop the transformation and ignore all following rows

import_set – the import set being transformed at that moment

log – function used for debugging the transformation (log.info(), log.error(), etc.)

map – information about the transform map record being processed

name – the name of the field on the target record that is currently being processed

source – A GlideRecord of the row being read in and processed

status_message – an output message sent in the XML response

target – A GlideRecord of the record being added to the target table

value – the display value of the field on the source record that is currently being processed

348) what is coalesce and it is options?

Answer: Coalesce is commonly used with Transform maps in ServiceNow. It is used to set a field as unique, so that when data is imported : - If ServiceNow identifies the data against an existing record using Coalesce, then it will update that record.

It will avoid duplicate records.

349) can we set coalesce for multiple mappings?

Answer: Yes, You can. If we keep coalesce on multiple fields, All coalesce fields should match to update record. if any one field is not matching, the record will be created.

350) what are the types of field maps in transform Map or how many ways we can map?

Answer:

Automap Matching fields

Mapping Assist

Script mapping

351) Does data policy work when you are importing the data through data sources?

Answer: Yes, It will work not only from form, it will from other data sources as well.

352) What is Import Set API?

Answer: The Import Set API allows for you to push data into an import set table and either asynchronously or synchronously transform it.

We will push the data into temporary tables first using Import Set API

353) what are the options available in transform map page and explain in details (run business rule.....etc)?

Answer: Run Business Rule, Run Script, Enforce mandatory fields, Copy empty fields, Create new record on empty coalesce fields

354) In transform script, what are ignore and error parameters?

Answer:

Error -- When set to true, aborts the current import action and logs an error message in the Import Set Log.

Ignore -- Specify the transformation action occurring on the target record. This value can be either "insert" or "update".

355) In transform script, what are source and target?

Answer:

Contains the import source record currently being transformed. Specify a specific field from the source record as an object property.

Contains the import target record currently being inserted. Specify a specific field from the target record as an object property.

356) how to stop 10th row transformation using transform script?

Answer: on the onBefore transform script -- You can use below line

```
if(source.row == 10){  
  ignore = true;  
}
```

357) how to transform only INSERTS while doing transforming the data?

Answer:

```
if(action == "update"){  
  ignore = true;  
}
```

358) what is a web service?

Answer: A web service is a set of open protocols and standards that allow data to be exchanged between different applications or systems.

359) what is end point?

Answer: Endpoints usually have a structured URL that includes the ServiceNow instance address and the API path.

ServiceNow provides a REST API for interacting with its platform. Each REST API has specific endpoints for different operations such as creating, reading, updating, or deleting records.

360) what request and response?

Answer:

A request is a message sent by one system to another to perform a specific operation or retrieve certain data. In ServiceNow integrations, this can involve REST or SOAP APIs, web services, or other communication protocols.

A response is the message sent back from the system that received the request. It contains the result of the operation requested, including any data or status information.

361) What is basic Authentication?

Answer: In Integration, We use username, password for basic authentication.

We create a User Account, We add specific roles to user account, we share user account details to external system.

362) how to test a API or end point?

Answer: Using REST API Explorer

363) what is postman and how you use it?

Answer: Postman is a software application that allows developers to test, document, and share APIs (Application Programming Interfaces). Developers widely use it to simplify the process of testing APIs by providing a user-friendly interface for making requests, viewing responses, and debugging issues.

364) what are rest methods and use of each?

Answer:

Rest API is an application program interface that uses below methods to perform above mentioned operations such as create, update, read and delete: Post (used to create information) Get (used to read information or resource) Put (used to update the information or resource)

365) What are the tools you integrated with ServiceNow?

Answer: HPSM, JIRA, Salesforce..Etc..

366) What is Scripted REST API?

Answer: A Scripted REST API in ServiceNow allows you to create a custom API endpoint within the ServiceNow platform, where you can define custom logic and data retrieval to expose specific functionalities or data.

367) what is rest message and Scripted REST API?

Answer:

Scripted REST API (ServiceNow):

Purpose: A Scripted REST API in ServiceNow allows you to create a custom API endpoint within the ServiceNow platform, where you can define custom logic and data retrieval to expose specific functionalities or data.

Customization: It is highly customizable and is typically used for creating custom APIs within the ServiceNow platform. You can define the structure of the response, perform data transformations, and even implement custom authentication and authorization logic.

Platform Integration: Scripted REST APIs are often used to interact with and manipulate data within the ServiceNow platform itself.

Example (ServiceNow): Suppose you are a ServiceNow administrator, and you want to create a custom API that returns a list of high-priority incidents along with additional information. You can create a Scripted REST API to define the structure of the response, apply custom filters, and perform any necessary data transformations. This API can then be accessed within ServiceNow or by external systems to retrieve this specific dataset.

REST Message (ServiceNow):

Purpose: In ServiceNow, a REST message is used to make outbound HTTP requests from within the platform to external web services or APIs. It is primarily used for integrating with external systems or services.

Communication: REST messages are not APIs themselves; instead, they are used to send HTTP requests (GET, POST, PUT, DELETE) to external web services or APIs and receive responses.

Integration: REST messages in ServiceNow are essential for connecting your ServiceNow instance to external systems, such as third-party APIs, and fetching data from those external sources.

Example (ServiceNow): Imagine you are using ServiceNow for IT service management, and you need to retrieve incident data from an external incident tracking system. In this scenario, you would create a REST message in ServiceNow to send an HTTP GET request to the external system's API endpoint, fetch the incident data, and then use this data within your ServiceNow instance to maintain an integrated view of incidents from both systems.

368) how to involve a rest method from script?

Answer:

```
var request = new sn_ws.RESTMessageV2();
    request.setEndpoint('https://api.ip2country.info/ip?' + address);
    request.setRequestHeader("Accept", "application/json");
    request.setRequestHeader('Content-Type', 'application/json');
    response = request.execute();
    var JSONData = new global.JSON().decode(response);
    gs.print('Country Code: ' + JSONData.countryCode + ' Country Name: ' + JSONData.countryName);
```

```
gs.log(response.getBody());
```

369) how to involve a soap message or soap method from script?

Answer:

```
var s = new sn_ws.SOAPMessageV2('<Soap Message Name>', '<Soap Message Function>');
```

```
//override authentication profile  
//authentication type ='basic'  
//r.setAuthentication(authentication type,profile name);
```

```
s.setStringParameterNoEscape('first_name', 'TestReqq');  
s.setStringParameterNoEscape('last_name', 'TestReqqLast');  
s.setStringParameterNoEscape('nbr', '4785');  
s.setStringParameterNoEscape('requesting_id', '<WebServiceName>');  
var response = s.execute();  
var responseBody = response.getBody();  
var status = response.getStatusCode();
```

370) What is API?

Answer: API integration creates a standardized, universal method for data exchange between applications. As such, it also gives businesses a clear, end-to-end view of all traffic within relevant systems and processes.

371) what is MID server and uses?

Answer: (MID) Server is a Java application that runs on a server on your local network. MID Servers facilitate communication and data movement between a single ServiceNow® instance and external applications, data sources, and services.

372) what are the common integration failures and how to find and resolve them in servicenow?

Answer:

Authentication failure -- Check user name and password

Bad Request -- Syntax error

Page not found -- Check the end point

Response format - Check the response format to solve the issues.

373) what is the difference between PUT and PATCH?

Answer: In the REST world, PUT and PATCH have different semantics. PUT means replace the entire resource with given data (so null out fields if they are not provided in the request), while PATCH means replace only specified fields. For the Table API, however, PUT and PATCH mean the same thing. PUT and PATCH modify only the fields specified in the request.

374) what is REST API Explorer?

Answer: ServiceNow's REST API Explorer is an application to construct and test API requests to a ServiceNow instance. The REST API Explorer is available to users with the rest_api_explorer role or the admin role. To open the REST API Explorer, use the All menu to open System Web Services > REST > REST API Explorer.

375) what is the difference Between SOAP and REST?

Answer: SOAP and REST are two different approaches to API design. The SOAP approach is highly structured and uses XML data format. REST is more flexible and allows applications to exchange data in multiple formats.

376) how much time the data will be available in import set tables. how to setup clean up of import set data for every two days?

Answer: By default, the job deletes all import sets that are older than 7 days, as well as the associated import set table records.
Navigate to All > System Import Sets > Import Set Tables > Scheduled Cleanup.
Set the interval to 2 days.

377) what are the type of authentications?

Answer: ServiceNow REST APIs use basic authentication, mutual authentication and OAuth to authorize user access to REST APIs/endpoints. You can also configure your instance to use multi-factor authentication to access REST APIs.

378) what is the oauth authentication?

Answer: It is token based authentication

For more details, Please go through below

<https://www.servicenow.com/community/developer-blog/oauth-2-0-with-inbound-rest/ba-p/2278926>

379) what is JSON.Stringfy()?

Answer: JSON.Stringfy() is used to convert object type values into strings.

380) which format the integrations will use to transfer the data?

Answer:json,xml,text/markdown,text/csv,text/html..Etc

381) what is integration and what is spoke?

Answer: A spoke is a scoped application that includes Flow Designer or Integration Hub actions or subflows. A spoke is a logical grouping of related actions, subflows, and supporting application files. Developers and Process Designers can reuse spoke logic in their own applications. IntegrationHub includes baseline spokes.

382) explain few actions in Flow Designer?:

Answer: Create Record, Update Record, Delete Record, Look Up Records, Send Email, Send Notification, get Catalog variables.

383) what are the return codes of REST end point?

Answer:

200 OK. The 200 OK status code indicates the request succeeded. ...

303 See Other. The 303 See Other status code indicates that you are being redirected to another resource via the "Location" response header. ...

400 Bad request. ...

404 Resource not found. ...

500 Internal server error.

384) what is 400 and 401 error codes?

Answer:400 Bad Request is the status code to return when the form of the client request is not as the API expects. 401 Unauthorized is the status code to return when the client provides no credentials or invalid credentials.

385) what is 200 ad 201 error codes?

Answer:200 OK: This status code indicates that the request was successful, and the server returned the requested data. 201 Created: This status code means that the request was successful, and the server created a new resource.

386) what are query parameters in rest message?

Answer: Request parameters are added to the endpoint URL by the REST API Explorer when the request is sent. The query parameters are specific to the selected API method. A default set of query parameters are displayed for the API.

387) what is the difference between query parameter and path parameters?

Answer:

Path Parameters: Path Parameters are used to include dynamic values directly in the URL path of a request. They are denoted by a specific placeholder within curly braces {} in the URL.

Endpoint: /api/users/{userId}

In this case, {userId} is a path parameter, and the actual user ID value would be provided in its place in the URL. For instance, if the user ID is "12345", the complete URL would be /api/users/12345.

Query Parameters are appended to the URL as key-value pairs, separated by an ampersand (&). They are used to provide additional information or parameters to the server. Query parameters do not impact the structure of the URL itself, and they can be optional.

Endpoint: /api/users?role=admin&active=true

In this case, the query parameters "role" and "active" are appended to the URL. The server can then use these parameters to filter and return the desired user list based on the criteria specified.

path parameters are used when you need to identify or retrieve a specific resource directly from the URL path, while query parameters are used for providing additional information or parameters to the server.

388)How to run scheduled job using script ?

Answer: SncTriggerSynchronizer.executeNow(current);

```
var rec = new GlideRecord('sysauto_script');
rec.get('name', 'YOUR_JOB_NAME_HERE');
if (typeof SncTriggerSynchronizer != 'undefined')
    SncTriggerSynchronizer.executeNow(rec);
else
```

```
Packages.com.snc.automation.TriggerSynchronizer.executeNow(rec);
```

389) How to trigger Catalog item using script?

Answer: Using Cart API

```
var cartId = GlideGuid.generate(null);
var cart = new Cart(cartId);
//add your requested item to the cart by sys_id of the catalog item
var item = cart.addItem('4054428fdb151f0097679ec6db9619c0', 1);

//fill in the variables on the request item form
cart.setVariable(item,"u_requested_for", "e80edd6edbec6bc097275e25ca9619a4");
cart.setVariable(item,"contact_number", "0");
cart.setVariable(item,"assignment_group", "87ec1c342b2e71406c487fb5a8da1524");
cart.setVariable(item,"application_service", "cdedfbcedb0f9744c291c170ba9619a7");
cart.setVariable(item,"short_description", email.subject);
cart.setVariable(item,"description", email.body_text);
var rc = cart.placeOrder();
```

390) How to trigger Flow Designer using script?

Answer: `sn_fd.FlowAPI.getRunner().flow('global.send_app_remainder').inForeground().run();`

391)How to trigger Flow Action using script?

Answer:

```
// Start Asynchronously: Uncomment to run in background. Code snippet will not have access to outputs.
// sn_fd.FlowAPI.getRunner().action('global.date_defference').inBackground().withInputs(inputs).run();

// Execute Synchronously: Run in foreground. Code snippet has access to outputs.
var result = sn_fd.FlowAPI.getRunner().action('global.date_defference').inForeground().withInputs(inputs).run();
var outputs = result.getOutputs();
```

392)How to trigger subflow using script?

Answer:

```
// Start Asynchronously: Uncomment to run in background. Code snippet will not have access to outputs.
```

```
// sn_fd.FlowAPI.getRunner().subflow('sn_grc_workspace.wait_for_state_change').inBackground().withInputs(inputs).run();
```

```
// Execute Synchronously: Run in foreground. Code snippet has access to outputs.
```

```
var result = sn_fd.FlowAPI.getRunner().subflow('sn_grc_workspace.wait_for_state_change').inForeground().withInputs(inputs).run();
```

```
var outputs = result.getOutputs();
```

393)How to trigger flow designer asynchronously?

Answer:

```
sn_fd.FlowAPI.getRunner().flow('global.send_app_remainder').inBackground().run();
```

394)How to trigger action asynchronously?

Answer:

```
// Start Asynchronously: Uncomment to run in background. Code snippet will not have access to outputs.
```

```
// sn_fd.FlowAPI.getRunner().action('global.date_defference').inBackground().withInputs(inputs).run();
```

```
// Execute Synchronously: Run in foreground. Code snippet has access to outputs.
```

```
var result = sn_fd.FlowAPI.getRunner().action('global.date_defference').inForeground().withInputs(inputs).run();
```

```
var outputs = result.getOutputs();
```

395)How to trigger subflow asynchronously?

Answer:

```
// Start Asynchronously: Uncomment to run in background. Code snippet will not have access to outputs.
```

```
// sn_fd.FlowAPI.getRunner().subflow('sn_grc_workspace.wait_for_state_change').inBackground().withInputs(inputs).run();
```

```
// Execute Synchronously: Run in foreground. Code snippet has access to outputs.
```

```
var result = sn_fd.FlowAPI.getRunner().subflow('sn_grc_workspace.wait_for_state_change').inForeground().withInputs(inputs).run();
```

```
var outputs = result.getOutputs();
```

396) How to restrict Assigned to values based on Assignment Group?

Answer: Using advanced Reference Qualifier.

397)How to access Request values in Scripted Rest API?

Answer: request.body.data.variable_name;

398) How to pass response from scripted REST API?

Answer: response.setBody(obj);

399) how to call the Rest Message in Server Side?

Answer: new sn_ws.RESTMessageV2('messageName', 'httpmethodname');

400) what is execute and executeAsync?

Answer:

execute is synchronous

executeAsync is asynchronous

401) Apart from ITSM, what other modules you are aware of?

Answer: I have got experience on Scoped Applications along with ITSM, I have created multiple catalog items and record producers in the scoped applications.

402) what is incident management and explain life cycle?

Answer: Incident management is a series of steps taken to identify, analyze, and resolve critical incidents which could lead to issues in an organization.

403) what is problem management and explain life cycle?

Answer: The goal of problem management is to eliminate recurring incidents, prevent future incidents, and minimize the impact of incidents which cannot be prevented.

404) what is change management and explain life cycle?

Answer: Change Management in ServiceNow involves a systematic approach to controlling the life cycle of all changes, facilitating beneficial changes with minimal disruption to IT services. ServiceNow has a change management tool to manage the change process effectively.

405) what is knowledge management and explain life cycle?

Answer: Knowledge management is the systematic process of gathering, organizing, and sharing an organization's information and expertise for easy access. Technical resources, instructional documentation, FAQ pages, etc.

406) what are the types of change management?

Answer:

Standard change

Normal Change

Emergency Change

407) What is relation between incident and problem?

Answer:

An incident is something that affects a service (so a user, system, etc) - something that impacts productivity. Generally it is the effect. A problem is the underlying cause of one or more incidents.

408) what is the difference between incident management vs major management process?

Answer: Incident management is out of the box solution, included in the main package.

Major incident management is additional plugin that you subscribe for.

Whether one incident is major or not is a decision, taken by the major incident manager. Of course all Major incidents have priority 1 - this is somehow natural. If something is breaking your normal business operations, then it must be with a high priority in your queue, right?

Please read a bit more for ITSM processes, some ITIL training or something that will bring more clarity in your mind about processes. Then you'll see that the tool is following the processes, and works as it is supposed to work.

409) Difference between Parent Incident and Child Incident in ServiceNow?

Answer: Parent Incidents cannot be merged. Parents can be added to Children even after the Parent has been resolved. If a Parent incident becomes unresolved; the children will remain resolved. Children may be unresolved manually.

410) How to get Section Names on the form?

Answer: Using GlideForm, You can get section names -- `g_form.getSectionNames()`;

Section names follows all lower case letters(Ex-- Notes) and Spaces will be replaced with '_'(Ex -- related_records)

412) How to hide 'New' button on related list?

Answer: Click on any column options -- Configure -- List control -- Omit New Button.

413) How to hide 'Edit' button on related list?

Answer: Click on any column options -- Configure -- List control -- Omit Edit Button.

414) How to hide New button in related list only to few members?

Answer: Click on any column options -- Configure -- List control -- Omit New condition

We have to set answer variable to true/false, parent object refers to parent record of the related list.

415) Can you write client side API in condition fields?

Answer: No, Condition field is a server side script, you can only write server side script and APIs

416) If one ACL is granting access and one ACL is restricting the access, What will be final result?

Answer: If at least one ACL is granting the access, The user get access.

417) what is scrum master role?

Answer: In Agile methodologies, particularly within Scrum, the Scrum Master serves as a key facilitator, coach, and guardian of Agile practices.

418) What are dictionary Overrides?

Answer: If a field is coming from parent table to child table, Along with the fields, field properties will also come to the child tables.

To override that properties on the child tables, we use Dictionary Overrides.

such as Mandatory, Readonly, default value, Display Value, Dependent Value, calculated Value, Attributes, Reference Qualifiers.

419) Dictionary Override example?

Answer: If I have to make assignment Group field mandatory on Incident form, First, We have to check assignment Group field is from which table, If the Assignment Group field is from parent table, We have to go for Dictionary Overrides, so that Assignment Group field will be mandatory only on incident form.

420) Can I make Assigned To field readonly on incident form?

Answer: No, assigned To field is coming to task table, If we make it readonly, it will become readonly on all the child tables of task table.

421) What are the Dictionary Properties we can override?

Answer: Mandatory, Readonly, default value, Display Value, Dependent Value, calculated Value, Attributes, Reference Qualifiers.

422) What are Data Look Up rules?

Answer: If the priority value should be populated from Impact and Urgency, We create Priority Look Up rule. after that we have to create a Look Up Definition between source table and the target Table.

423) How to open incident from the URL?

Answer: [https://dev65206.service-now.com\(Base URL\)/+incident.do?sys_id=-1](https://dev65206.service-now.com(Base URL)/+incident.do?sys_id=-1)

424) How to open Incident List using short cut?

Answer: [https://dev65206.service-now.com\(Base URL\)/+incident_list.do](https://dev65206.service-now.com(Base URL)/+incident_list.do)

425) Short Cut to open the forms?

Answer: table_name.do

Example: incident.do, incident.FORM, incident.form

426) Short Cut to open List?

Answer: table_name.list

Example: incident.list, incident.LIST

428) Short cut to open the Filter?

Answer: table_name.filter

Example: incident.filter, incident.FILTER

429) Short Cut to open the Configurations?

Answer: table_name.config

Example: incident.config, incident.CONFIG

430) How to create Application Menu?

Answer: Application Navigater --> Unders System Definition --> Application Memus

431) What are the link types in the Modules?

Answer: New Record, List of Records, Separater, URL Arguments, Script Arguments, Report ...Etc...

432) What is Order in Modules?

Answer: Order means sequence

Order with lowest number will appear on the top.

Order with highest number will appear on the bottom.

433) What is Class field in Table?

Answer: Class field is created on table(Task) when it is extended by another table(incident, problem, Change...Etc), We can use class field to separate the child table data from parent table data.

434) What is Collection field on table?

Answer: When a table is created, Collection Type field will get created, In the collection type field we can add attribute to hide attachment, to capture the data in update sets, To start the auditing on the records..Etc..

435) How to remove or add fields from Preview Record Icon?

Answer: Base URL + table_name.do?sysparm_view=cxs_popup

We have to modify cxs_popup view.

436) What is the difference between catalog client Script and Table client Scripts?

Answer: We have 4 types of client Scripts on tables, We have only 3 types of client scripts in catalog client scripts. There is no oncelledit client script in catalog client scripts.

437) Client Scripts table name?

Answer:sys_script_client

438) Catalog Client scripts table name?

Answer:catalog_script_client

439) Business Rules table name?

Answer: sys_script

440) Ui policy table name?

Answer: sys_ui_policy,sys_ui_policy_action

441) UI Action table name?

Answer: sys_ui_action

442) UI Pages table name?

Answer: sys_ui_page

443) Script Execution order in ServiceNow?

Answer:

Query Business Rules
Display Business Rules
onload Client Script
onload UI policy
onchange Client script
onchange UI policy
onsubmit client script
Client UI Actions
Server Side UI Actions

Before business rules: Scripts configured to execute before the database operation with an order less than 1000.

Before engines. The following are not executed in any specific order:

Approval engine (for task and sys_approval_approver tables)

Assignment rules engine (for task tables)

Data policy engine

Escalation engine

Field normalization engine

Role engine - keeps role changes in sync with sys_user_has_role table (for sys_user, sys_user_group, sys_user_grmember, and sys_user_role tables)

Execution plan engine (for task tables)

Update version engine - creates version entry when sys_update_xml entry is written (for sys_update_xml table)

Workflow engine (for default workflows)

Before business rules: Scripts configured to execute before the database operation with an order greater than or equal to 1000.

The data base operation (insert, update, delete).

After business rules: Scripts configured to execute after the database operation with an order less than 1000.

After engines. The following are not executed in any specific order:

Label engine

Listener engine

Table notifications engine

Role engine - keeps role changes in sync with sys_user_has_role table (for sys_user, sys_user_group, sys_user_grmember and sys_user_role tables)

Text indexing engine

Update sync engine

Data lookup engine inserts or updates

Workflow engine (for deferred workflows)

Trigger engine (for all Flow Designer flows)

Email notifications. The following are executed based on the weight of the notification record:

Notifications sent on an insert, update, or delete

Event-based notifications

After business rules. Scripts configured to execute after the database operation with an order greater than or equal to 1000.

444) What is template in Email Script?

Answer: template is used to pass HTML content from Email script to Email Notification

445) Where do we track sent or received emails in ServiceNow?

Answer: Emails under System logs.

446) What are the debugging methods in script?

Answer: gs.log, gs.info, gs.warn.

447) How to create approval using script?

Answer:

```
var approval_gr = new GlideRecord('sysapproval_approver');
approval_gr.initialize();
approval_gr.state = 'requested';
approval_gr.approver = gs.getUserID();
approval_gr.sysapproval = current.sys_id;
approval_gr.insert();
```

448) How to wait for 2 members approvals in workflow?

Answer:

In Approval activiy, We have to choose Condition based on script

Use below line:

```
if(counts.approvaed == 2){
answer = 'approved'
}
```

```
if(counts.rejected == 2){
answer = 'rejected'
}
```

449) How to access catalog variables in workflow ?

Answer: `current.variables.variable_name`

450) How to set catalog task values using script?

Answer: using task object

```
task.short_description = 'Test value';
```

451) How to pass values to one activity to another activity in servicenow?

Answer: `workflow.scratchpad.variable_name;`

452) How to pass values from one action to another action in servicenow?

Answer: We have to create a flow variable.

We can set flow variables using data pills.

JavaScript:

1) Difference between “ == “ and “ === “ operators.?

Both are comparison operators. The difference between both the operators is that “==” is used to compare values whereas, “=== “ is used to compare both values and types.

2) Sort an array containing numbers using a 'for' loop?

Answer:

```
var Arr = [1, 7, 2, 8, 3, 4, 5, 0, 9];
```

```
for (var i = 1; i < Arr.length; i++)  
  for (var j = 0; j < i; j++)  
    if (Arr[i] < Arr[j]) {  
      var x = Arr[i];  
      Arr[i] = Arr[j];  
      Arr[j] = x;  
    }
```

```
console.log(Arr);
```

3) How to remove duplicate in array using for loop?

Answer: function getUnique(arr){

```
  let uniqueArr = [];
```

```
  // loop through array  
  for(let i of arr) {  
    if(uniqueArr.indexOf(i) === -1) {  
      uniqueArr.push(i);  
    }  
  }
```

```
    }  
  }  
  console.log(uniqueArr);  
}
```

```
const array = [1, 2, 3, 2, 3];
```

```
const array = [1, 2, 3, 2, 3];
```

```
// calling the function  
// passing array argument  
getUnique(array);
```

4) How to reverse an array?

Answer:

```
var array = [1, 2, 3, 4]
```

```
const reversedArray = []
```

```
for(var i = array.length - 1; i >= 0; i--) {  
  const valueAtIndex = array[i]
```

```
  reversedArray.push(valueAtIndex)  
}
```

```
console.log(reversedArray)
```

5) How to reverse a string?

Answer:

```
function reverseString(str) {
```

```
  // Step 1. Use the split() method to return a new array  
  var splitString = str.split(""); // var splitString = "hello".split("");  
  // ["h", "e", "l", "l", "o"]
```

```
  // Step 2. Use the reverse() method to reverse the new created array  
  var reverseArray = splitString.reverse(); // var reverseArray = ["h", "e", "l", "l", "o"].reverse();
```

```
// ["o", "l", "l", "e", "h"]

// Step 3. Use the join() method to join all elements of the array into a string
var joinArray = reverseArray.join(""); // var joinArray = ["o", "l", "l", "e", "h"].join("");
// "olleh"

//Step 4. Return the reversed string
return joinArray; // "olleh"
}
```

```
reverseString("hello");
```

6) How to check if a word is palindrome or not?

Answer:

```
function isPalindrome(str) {
    let j = str.length - 1
    for (let i = 0; i < str.length / 2; i++) {
        if (str[i] !== str[j]) {
            return false;
        }
        j--;
    }
    return true;
}
```

```
let str1 = "racecar";
let str2 = "nitin";
let str3 = "Rama";
```

```
console.log(isPalindrome(str1));
console.log(isPalindrome(str2));
console.log(isPalindrome(str3));
```

7) How to print Fibonacci sequence numbers?

Answer:

```
function fibonacci(num)
{
    var num1 = 0;
    var num2 = 1;
    var sum;
    var i = 0;
    for (i = 0; i < num; i++)
    {
        sum = num1 + num2;
        num1 = num2;
        num2 = sum;
    }
    return num2;
}
document.write(fibonacci(1));
document.write(fibonacci(12));
```

8) How many data types in javascript?

Answer: number, Boolean, string, null, undefined, object.

9) Some of the string methods?

Answer: toLowerCase(), toUpperCase(), trim(), indexOf(), replace(), replaceAll()..Etc..

10) How to convert string into object?

Answer: JSON.parse(str);

11) How to convert object into string?

Answer: JSON.stringify(obj);

